

The pathwidth complex

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Abstract

The minimization of open stacks problem (MOSP) asks for the order in which to make a set of patterns so that the fewest customer orders are open at once. [Linhares and Yanasse \(2002\)](#) listed twelve problems from operations research, VLSI design and graph theory, MOSP among them, as equal to it “plus or minus one”, and gave no proofs. We ask which of these equalities are true, and what follows when results, algorithms and benchmarks are moved between the twelve problems through pathwidth, the quantity that connects them. We find four things. First, eight of the twelve rows are exact, counting pathwidth itself: each equals pathwidth or pathwidth plus one. Two lie in bands one and two wide, and two are false as stated; every relation is proved, and machine-checked in Lean. Second, the literature has settled on the name pathwidth, used more than three times as often as the other eleven names together, and the three communities rarely cite one another. Third, the best exact algorithm for MOSP, the search of [Chu and Stuckey \(2009\)](#), is a pathwidth algorithm, and two of its published dominance theorems are false. The correct rule was already known for pathwidth; we show that it is tested by one bipartite matching per candidate, and prove the repaired search sound. In our implementation it costs at most 2% more search nodes on each benchmark set, and it re-proved all 115 benchmark optima that rested on the published rules, changing none. Fourth, we release 17,714 benchmark graphs, distinct up to isomorphism, from twenty collections, 16,087 of them with a certified pathwidth and a layout that attains it.

1 Introduction

A factory makes a set of patterns, one at a time, and each customer’s order needs some of them. A customer’s stack opens when the first pattern the customer needs is made, and it closes when the last one is made. The *minimization of open stacks problem* (MOSP) asks for the order of the patterns that keeps the fewest stacks open at any one time. In the cutting industry, where the problem arose, the patterns are cutting patterns and the customers are piece types ([Yanasse, 1997b](#)). Figure 1.1(a) gives a small instance. Six customers, a to f , need four patterns: $P_1 = \{a, b\}$, $P_2 = \{b, c, d\}$, $P_3 = \{c, d, e\}$ and $P_4 = \{e, f\}$. Made in order A, $P_1 P_2 P_3 P_4$, the patterns keep at most 3 stacks open. Made in order B, $P_1 P_3 P_4 P_2$, they keep 5 open, because customer b is served first but waits for P_2 , which comes last. Of the 24 orders, 2 reach 3 (order A and its reverse), 16 give 4 and 6 give 5. MOSP is NP-hard ([Linhares, 2001](#); [Linhares and Yanasse, 2002](#)).

The same matrix is a problem in VLSI design. In a *gate matrix layout* the patterns are gates, drawn as vertical lines in some order, and the customers are nets. Each net is a horizontal wire from the first gate it connects to the last, and wires that overlap must lie on different horizontal tracks. The task is to order the gates so that the fewest tracks suffice. Figure 1.1(b) and (c) draw the two orders. The wires are the same; only the gate order changes which wires overlap. Order A packs into 3 tracks and order B needs 5. The number of tracks a gate order needs is its *density*, the most wires crossing any one gate, and that is exactly the number of open stacks there. So the two problems compute the same number from the same matrix ([Linhares and Yanasse, 2002](#), Proposition 2).

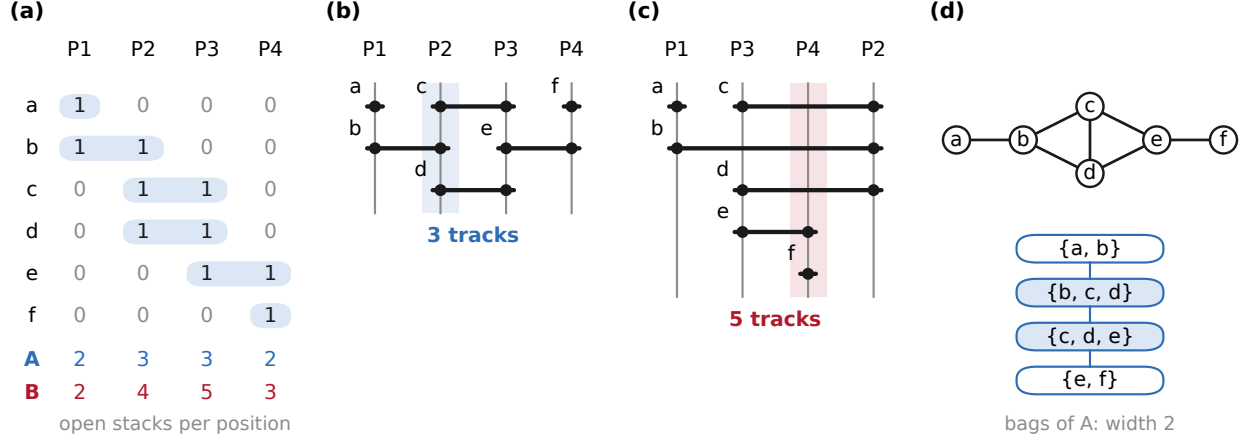


Figure 1.1: One instance, three readings, one number. (a) Six customers and four patterns, with each customer’s open stack shaded in order A; order A keeps at most 3 stacks open and order B at most 5, and only 2 of the 24 orders reach 3. (b, c) The same matrix as a gate matrix layout: the same two orders need 3 and 5 tracks (the shaded gate is where the density peaks). (d) The same matrix as a graph, one clique per pattern: the open sets of order A are the bags of a path decomposition of width 2, so the optimum is the pathwidth plus one (Theorem 2.3).

The same matrix is also a graph. Make each customer a vertex, and join two customers when some pattern needs both, so that each pattern becomes a clique (Figure 1.1(d)). Write down, for order A, the set of open stacks at each position: $\{a, b\}$, $\{b, c, d\}$, $\{c, d, e\}$, $\{e, f\}$. These sets cover every vertex and every edge, and each customer appears in a run of consecutive sets. In graph theory such a sequence is a *path decomposition*, its sets are *bags*, and its width is the size of its largest bag minus one. The least width over all path decompositions is the *pathwidth* of the graph, here 2. The construction works in both directions, so the fewest open stacks is always the pathwidth plus one (Theorem 2.3).

Linhaires and Yanasse (2002) collected twelve such problems, from operations research, VLSI design and graph theory, in the table that Table 1.1 reproduces. Each computes, on corresponding inputs, “a function $f(\Pi)$ that is either equal to the number of open stacks or closely related to it (plus or minus one)” (Linhaires and Yanasse, 2002, p. 1764). The table gives a reference for each problem, but no proofs and no graph of which problem reduces to which. “Plus or minus one” has two readings. Under the *literal* reading, f is within one of the number of open stacks on every input. Under the *fixed-offset* reading, f equals the number of open stacks plus one constant, the same on every input. Only the second makes solving one problem the same as solving another. We call the twelve *the pathwidth complex*, because pathwidth is both the name the literature uses most (Section 3) and the quantity through which every true relation in the table is proved (Section 2). This paper asks: *which of the equalities in the Linhares–Yanasse table are true, and what follows when results, algorithms and benchmarks are moved between the twelve problems through pathwidth?*

The table has a history that explains why the question is still open. It was first printed as Table 2.1 of the first author’s doctoral thesis (Linhaires, 2001, p. 13), supervised by the second author, which proved MOSP NP-hard and equal to gate matrix layout. Its rows were collected over the years of the doctorate, while reading the literature of each field. Each came with a reference and nothing else. No row’s relation to open stacks was checked, no published proof was restated, and no graph said which problem reduces to which. The table appeared again, row for row, as Table 1 of Linhares and Yanasse (2002), and it has been read since as a statement of equivalence. Garcia de la Banda and Stuckey (2007), for instance, credit the speed of their search in this journal to “the fact that the problem is fixed parameter tractable (Linhaires and Yanasse 2002)”. So the table is already used to justify algorithms, and whether it is true matters. If it is true, every result about any one of the twelve problems transfers to all the others. Looking back, the table was the thesis’s most important finding, and this article does what the thesis did not: it states each problem precisely, proves

Table 1.1: The Linhares–Yanasse table, reproduced from Linhares and Yanasse (2002, Table 1, p. 1764) with its row order, wording and references as printed, and its caption: “Display of a family of equivalent problems studied independently on the literature”. The shaded column is ours. It gives the relation to pathwidth (pw) that is true, proved in Section 2 (Table 2.2). Eight rows are exact, counting the pathwidth row itself; two are bands; two are false as the table states them.

Problem	Discipline	References	This paper
MOSP	Operations research	Yanasse (1997b); Fink and Voß (1999)	exact: $\text{pw} + 1$
Gate matrix layout	VLSI design	Möhring (1990); Wing et al. (1985)	exact: $\text{pw} + 1$
One-dimensional logic	VLSI design	Ohtsuki et al. (1979)	exact: $\text{pw} + 1$
PLA folding	VLSI design	Möhring (1990)	false ^a
Interval thickness	Graph theory	Kashiwabara and Fujisawa (1979)	exact: $\text{pw} + 1$
Node search game	Graph theory	Kirousis and Papadimitriou (1985)	exact: $\text{pw} + 1$
Edge search game	Graph theory	Kirousis and Papadimitriou (1986)	band: pw to $\text{pw} + 2$
Narrowness	Graph theory	Kornai and Tuza (1992)	exact: $\text{pw} + 1$
Split bandwidth	Graph theory	Fomin (1998)	band: pw to $\text{pw} + 1$
Graph path-width	Graph theory	Kinnersley (1992)	exact: pw (reference)
Edge separation	Graph theory	Lengauer (1981)	false ^b
Vertex separation	Graph theory	Kinnersley (1992)	exact: pw

^aExact ($\text{pw} + 1$) for multiple folding, with any number of nets per track. ^bAs cutwidth; the vertex separator game of the same reference is exact (pw , for graphs with an edge).

what is true and refutes what is not. The communities that could have checked the table rarely read each other: of 844 works that cite the table’s papers, only six cite both an open-stacks paper and a graph-theory paper (Section 3).

What we find.

- *The table, settled.* Eight of the twelve rows are exact, the pathwidth row itself counted as one of them: the other seven equal pathwidth or pathwidth plus one on every input, outside degenerate cases such as an empty graph. Two rows, split bandwidth and edge search, lie in bands one and two wide. Two rows, PLA folding and edge separation, are false as stated, with counterexamples. A variant of each false row is exact (Section 2).
- *One name.* The literature has settled on “pathwidth”, which has more than three times as many relevant works as the other eleven names together. Besides pathwidth, the most active community over 2005–2024 is the open-stacks one, and the exact search of Section 4 came from it (Section 3).
- *A repaired search.* The best exact algorithm for MOSP, the search of Chu and Stuckey (2009), is also a pathwidth algorithm. Two of its published dominance theorems are false, as are their later restatements (Chu, 2011; Fink, 2012). The correct rule was already known in the exact pathwidth literature, and we show that it can be tested by one bipartite matching per candidate. The repaired search is proved sound. Over each benchmark set it changes the total number of search nodes by at most 2%, and no wrong answer of the published rules on a whole instance is known (Section 4).
- *Benchmarks re-certified, and a dataset.* The repaired search re-proved all 115 benchmark optima that rested on the published rules, and none changed (as of 6 October 2026). We release 17,714 graphs, distinct up to isomorphism, from twenty benchmark collections of the twelve problems; 16,087 have a certified pathwidth with a layout that attains it, and so a certified value for every exact problem in the table (Section 5).

How to read the paper. Section 2 states the twelve problems and proves what is true. Section 3 measures which names and communities the literature uses. Section 4 reads the exact open-stacks search as a pathwidth algorithm, finds the two false rules and proves the repaired search sound. Section 5 reports its cost and the dataset, and Section 6 answers the question above. Every theorem we state is machine-checked in the Lean 4 proof assistant. The reader need not read Lean: the proofs are given in the text, and Appendix A maps each result to its formal statement. Appendix B holds the supporting material, Appendix C the bibliometric method, and the data, code and proofs are described at the end of the paper (page 24).

2 The equivalences, proved

This section answers the first half of our question: which rows of the Linhares–Yanasse table are true, and in what sense. Section 1 separated two readings of “plus or minus one”. We call a row *exact* when it meets the fixed-offset reading: its value is pathwidth, or pathwidth plus one, on every input. We call it a *band* when it meets only the literal reading: its value stays within a fixed distance of pathwidth, and both ends of that range occur. We call it *false* when it meets neither. The answer is that eight rows are exact, two are bands and two are false. Section 2.1 states the problems. Section 2.2 proves the answer in four steps: the hub theorem, which ties MOSP to pathwidth; the other exact rows; the bands; and the false rows. Table 2.2 and Figure 2.1 summarise it.

All theorems below are verified in the Lean proof assistant, unless the text says otherwise; Appendix A lists the formal statements. Where a result is published, we cite the published proof and do not repeat it.

Notation. Graphs are finite and simple, $G = (V, E)$, $n = |V|$. A *layout* is a bijection $L : V \rightarrow \{1, \dots, n\}$. For a 0/1 matrix M with rows R (customers, nets, piece types) and columns C (patterns, gates, products), the *MOSP graph* G_M has vertex set R , with $r \sim r'$ when some column has a 1 in both rows; each column becomes a clique on its rows. For an order π of the columns, row r is *active* at position j if it has 1s in columns x, y with $\pi(x) \leq j \leq \pi(y)$.

2.1 The problems

Each problem is a decision problem with an instance and an integer k ; its value is the least k for which the answer is yes. Four of the twelve carry the rest of the paper, and we define them in full. Figure 1.1 shows each on one instance.

Pathwidth $\text{pw}(G)$. Is there a sequence of bags $X_1, \dots, X_r \subseteq V$ covering V , with every edge inside some bag, $X_i \cap X_\ell \subseteq X_j$ whenever $i \leq j \leq \ell$, and $|X_i| \leq k + 1$? (Kinnersley, 1992, p. 346). The condition on $X_i \cap X_\ell$ is the *interval property*: the bags that contain a vertex are consecutive.

Vertex separation $\text{vs}(G)$. Is there a layout L such that for every $i < n$ at most k vertices u with $L(u) \leq i$ have a neighbour v with $L(v) > i$? (Kinnersley, 1992, p. 346).

MOSP $Z(M)$. Given a 0/1 matrix M of customers by patterns, is there an order of the patterns with at most k customers active at every position? (Linhares and Yanasse, 2002, eqs. (1)–(2)).

Gate matrix layout $t(M)$. Given a net–gate matrix, is there an order of the gates and an assignment of nets to k tracks such that nets active at a common position get different tracks? (Möhring, 1990, p. 18); (Wing et al., 1985, Problem 1).

The other eight problems are needed only to read the table. Table 2.1 gives each in one line; Appendix B.1 gives the formal definitions, with the sources.

Table 2.1: The other eight problems of the Linhares–Yanasse table, in one line each. Each value is the least k that is possible. Formal definitions are in Appendix B.1.

Problem	In one line
One-dimensional logic	Order the gates of a circuit so that the intervals spanned by the nets overlap at most k at any point. Its graph is the <i>net graph</i> H , which joins two nets that share a gate. A boundary variant pins two gates to the ends.
PLA folding $\text{pla}(M)$	Gate matrix layout in at most k tracks, with at most two nets on a track. With no limit per track (<i>multiple folding</i>) it is gate matrix layout itself.
Interval thickness $\theta(G)$	Add edges to G to make an interval graph whose largest clique has at most k vertices.
Node search $\text{ns}(G)$	Searchers are placed on and removed from vertices. An edge is cleared when both its ends hold a searcher, and becomes contaminated again if a searcher-free path joins it to a contaminated edge. Clear every edge with at most k searchers at once. $\text{mns}(G)$ is the same with no recontamination allowed.
Edge search $\text{es}(G)$	Node search with a third move: sliding a searcher along an edge clears it. $\text{pes}(G)$ is the same over <i>progressive</i> strategies, which never recontaminate an edge.
Narrowness $\nu(G)$	Add the vertices one at a time in some order, and drop each as soon as all its neighbours have been added. Never hold more than k .
Split bandwidth $\text{sb}(G)$	Repeatedly split a vertex into an edge, sharing its neighbours between the two ends, so that the result has a layout in which adjacent vertices are at most k apart (its <i>bandwidth</i>).
Edge separation	The table’s reference (Lengauer, 1981) supports three readings: cutwidth $\text{cw}(G)$, the most edges crossing a gap of a layout; modified cutwidth $\text{mcw}(G)$, the most edges passing strictly over a vertex; and Lengauer’s vertex separator game $\text{VSG}(G)$, a pebbling game that counts the unpebbled vertices adjacent to pebbled ones.

2.2 What is true

Table 2.2 gives the answer row by row, and Figure 2.1 draws it as a graph around pathwidth. The rest of this subsection proves it.

The hub. Two theorems tie the table to pathwidth. The first joins the two graph rows, vertex separation and pathwidth. The second joins MOSP to pathwidth, and through it all four matrix problems. The second rests on one lemma about path decompositions. For a vertex v , let $\text{first}(v)$ and $\text{last}(v)$ be the least and greatest indices of bags containing v ; by the interval property, $v \in X_i$ exactly when $\text{first}(v) \leq i \leq \text{last}(v)$.

Lemma 2.1 (cliques sit in a bag). *In a path decomposition of G , every clique of G lies inside one bag (Appendix A, A.1).*

Proof. The bags containing a vertex u of the clique form the interval $[\text{first}(u), \text{last}(u)]$. Two vertices of the clique are adjacent, so their edge lies in a bag and their intervals meet. Pairwise intersecting intervals of a line have a common point (the Helly property): let a be the largest left end; every right end is at least every left end, so at least a , and a lies in all of them. So all the intervals share an index i , and the clique lies in X_i . $\square$

The first theorem is known. It says the two graph rows of the table are the same quantity.

Theorem 2.2 (vertex separation is pathwidth; known). *For every graph, $\text{vs}(G) = \text{pw}(G)$ (Appendix A, A.2).*

This is Theorem 3.1 of Kinnersley (1992). Both directions are constructions. From a layout of separation k , each bag is the set counted at a cut together with the next vertex. From a decomposition, ordering the vertices by first gives a layout of the same width.

Table 2.2: The Linhares–Yanasse table, settled. Eight rows are exact, two are bands and two are false, and every true relation is a relation to pathwidth. *Relation*: what is true and proved in Lean, on the graph the problem lives on (the MOSP graph or the net graph H for the matrix problems, the input graph otherwise). *Published source*: the published work the row rests on; the Lean proofs are listed in Appendix A. *Status*: *exact*, a fixed offset; *band*, an inequality whose ends both occur; *false*, the table’s claim fails, with a counterexample family proved in Lean.

Problem	Published source	Relation	Status
MOSP	Yanasse (1997a); Fellows and Langston (1989); Linhares and Yanasse (2002)	$Z(M) = \text{pw}(G_M) + 1$ (M has a 1)	exact
Gate matrix layout	Möhring (1990)	$t(M) = Z(M) = \text{pw}(G_M) + 1$	exact
One-dim. logic	Ohtsuki et al. (1979)	$\text{tracks} = \theta(H) = \text{pw}(H) + 1$; boundary variant $\geq \text{pw}(H) + 1$	exact
PLA folding	Möhring (1990)	$\max(\text{pw} + 1, \lceil N /2 \rceil) \leq \text{pla}$, gap unbounded; multiple folding $= \text{pw} + 1$	false; variant exact
Interval thickness	Möhring (1990); Kirousis and Papadimitriou (1985)	$\theta = \text{pw} + 1$ ($V \neq \emptyset$)	exact
Node search	Kirousis and Papadimitriou (1985, 1986); Bienstock and Seymour (1991)	$\text{ns} = \text{mns} = \text{vs} + 1 = \text{pw} + 1 = \theta$ ($E \neq \emptyset$)	exact
Edge search	Ellis et al. (1994); Kirousis and Papadimitriou (1986)	$\text{vs} \leq \text{es} \leq \text{vs} + 2$, also for pes	band
Narrowness	Kornai and Tuza (1992)	$\text{v} = \text{pw} + 1$ ($V \neq \emptyset$)	exact
Split bandwidth	Fomin (1998)	$\text{pw} \leq \text{sb} \leq \text{pw} + 1$	band
Path-width	Kinnersley (1992)	the reference	exact
Edge separation ^a	Lengauer (1981)	cw, mcw unbounded against pw (stars); $\text{VSG} = \max(1, \text{vs})$	false; VSG exact
Vertex separation	Kinnersley (1992)	$\text{vs} = \text{pw}$	exact

^aAlso misattributed: what Lengauer (1981) proves exactly is the vertex separator game, which is the table’s vertex separation row.

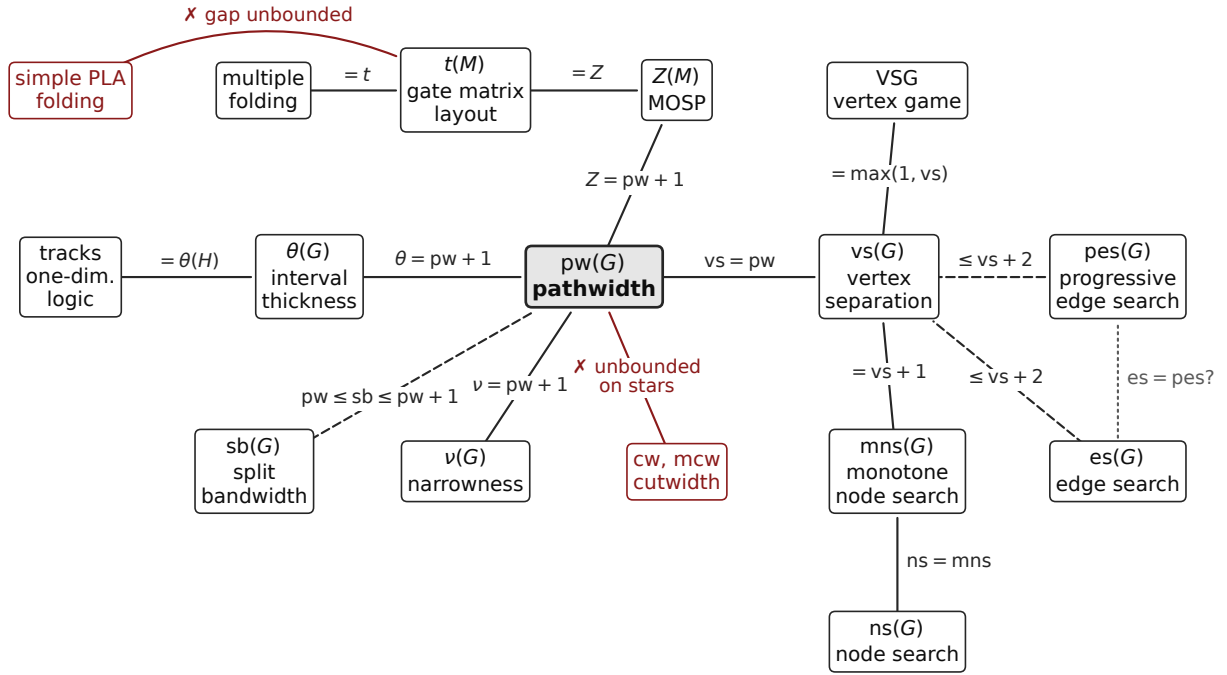


Figure 2.1: Pathwidth is the hub of the table: every exact relation is one or two steps from it, and the only failures are the two false claims, in dark red. Solid edges are exact relations and dashed edges bands, both proved in Lean; the dotted edge is LaPaugh's $\text{es} = \text{pes}$, the one relation not proved, which no row needs. Each false claim is refuted by a counterexample family proved in Lean. The four matrix problems (top left) live on the MOSP graph G_M or the net graph H . Hypotheses are omitted: M has a 1, $V \neq \emptyset$, or G has an edge, as in Table 2.2. Two edges that follow from the others, $\text{mns} = \theta$ and $\text{ns} = \text{vs} + 1$, are not drawn.

The second theorem is the one the rest of the paper uses. In Figure 1.1(d), the sets of customers active at each step of the good order are the bags of a path decomposition of the MOSP graph. The theorem says that this always works, in both directions.

Theorem 2.3 (MOSP is pathwidth plus one). *If M has at least one 1, then $Z(M) = \text{pw}(G_M) + 1$ (Appendix A, A.3).*

Proof. ($\leq$) Take a decomposition of G_M of width k . The rows of each column form a clique, so by Lemma 2.1 they lie in a bag $X_{b(c)}$. Order the columns by $b(c)$. If row r is active at the position of column c , it has columns x, y with $b(x) \leq b(c) \leq b(y)$, and $r \in X_{b(x)} \cap X_{b(y)}$, so the interval property gives $r \in X_{b(c)}$. At most $k + 1$ rows are active there.

($\geq$) Take an order with at most Z active rows at every position. The active sets, with a singleton bag for each row with no 1, form a path decomposition: rows sharing a column are both active at its position, and a row is active on an interval of positions. Its width is $Z - 1$, using $Z \geq 1$. $\square$

For the table, this settles the MOSP row as exact, with offset one, on the MOSP graph. Read as a cutting problem, Theorem 2.3 is Proposition 5 of Yanasse (1997a), which turns each pattern into a clique on its piece types. Read as a layout problem, it is Theorem 7 of Fellows and Langston (1989), with the column-expansion Lemma 4.1 of Fellows and Langston (1987). Linhares and Yanasse (2002, Prop. 2) identify MOSP with gate matrix layout. The proof above needs no assumption that the instance is reduced.

The graph matters. The MOSP literature also uses a second graph, with the patterns as vertices, joined when they share a customer, to split an instance into independent parts (Yanasse and Senne, 2010). It is natural to expect its pathwidth to measure the open stacks as well; it does not, in either direction. One pattern shared by four customers who each have a private pattern has $Z = 4$, while its pattern graph is the star $K_{1,4}$, of pathwidth 1 (Appendix A, A.4). A single customer with many patterns errs the other way.

The exact core. The other exact rows attach to Theorems 2.2 and 2.3 by published arguments. We do not reprove them on paper; each item below cites the published proof, and the Lean proofs are listed in Appendix A. We call the problems that equal pathwidth or pathwidth plus one on every input the *exact core*: the eight exact rows, with multiple folding and the vertex separator game.

Theorem 2.4 (the exact core).

1. *Gate matrix layout is MOSP on the same matrix: if M has a 1, $t(M) = Z(M) = \text{pw}(G_M) + 1$. For a fixed gate order the fewest tracks equals the most nets active at one position, by the left-edge algorithm (Möhring, 1990, p. 31) (Appendix A, A.5).*
2. *Interval thickness: if $V \neq \emptyset$, $\theta(G) = \text{pw}(G) + 1$ (Möhring, 1990, Prop. 3.5) (Appendix A, A.6).*
3. *One-dimensional logic: if every net lies on a gate, the fewest tracks is $\theta(H)$, and if some gate connects a net it is $\text{pw}(H) + 1$ (Ohtsuki et al., 1979, Thm 3) (Appendix A, A.7).*
4. *Narrowness: if $V \neq \emptyset$, $v(G) = \text{pw}(G) + 1$, and for each order σ , $v(\sigma) = \text{vs}(\sigma^{\text{rev}}) + 1$ (Kornai and Tuza, 1992, Prop. 3.1) (Appendix A, A.8).*
5. *Node search: for every graph $\text{ns}(G) = \text{mns}(G)$, and if $E \neq \emptyset$, $\text{ns}(G) = \text{vs}(G) + 1 = \text{pw}(G) + 1 = \theta(G)$ (Kirosis and Papadimitriou, 1985, Thm); (Kirosis and Papadimitriou, 1986, Thms 2.3, 4.1) (Appendix A, A.9).*
6. *Multiple folding: the fewest tracks is $t(M) = \text{pw}(G_M) + 1$ (Möhring, 1990, Thm 3.14) (Appendix A, A.10).*

7. *Lengauer’s vertex separator game*: $\text{VSG}(G) = \max(1, \text{vs}(G))$ ([Lengauer, 1981, Thm 4](#)) ([Appendix A, A.11](#)).

For the table, this makes the gate matrix layout, one-dimensional logic, interval thickness, node search and narrowness rows exact, and gives an exact variant of each false row. A single certified pathwidth therefore answers every problem in the exact core; [Section 5.4](#) uses this.

Bands. Two rows are not fixed offsets of pathwidth, but stay within a fixed distance of it.

Theorem 2.5 (bands). *For every graph, $\text{pw}(G) \leq \text{sb}(G) \leq \text{pw}(G) + 1$ ([Fomin, 1998, Thm 8](#)), and $\text{vs}(G) \leq \text{es}(G) \leq \text{vs}(G) + 2$, and the same for pes ([Ellis et al., 1994, Thm 2.1](#)) ([Appendix A, A.12](#)).*

Every end of both bands occurs. K_2 has $\text{sb} = \text{pw} = 1$ and $\text{es} = \text{vs} = 1$; the star $K_{1,3}$ has $\text{pw} = 1$ and $\text{sb} = 2$, and $\text{vs} = 1$ with $\text{es} = 2$; and $K_{3,3}$ has $\text{vs} = 3$ with $\text{es} = 5$. These values are computed by brute force from each quantity’s definition, not proved in Lean.¹ [Fomin \(1998\)](#) assumes a connected graph with at least two vertices; the Lean proof needs neither. LaPaugh’s equality $\text{es} = \text{pes}$ ([LaPaugh, 1993](#)) is the one relation we state and do not prove, and no row needs it.

For the table, this means the following. Relative to the number of open stacks $Z = \text{pw} + 1$, split bandwidth lies in $[Z - 1, Z]$ and edge search in $[Z - 1, Z + 1]$. Both stay within one of Z , so they meet the literal reading of “plus or minus one”. Since both ends occur, neither is a fixed offset, so neither meets the fixed-offset reading.

False rows. Two rows fail even the literal reading: no constant bounds their distance from pathwidth.

Theorem 2.6 (two rows are false).

1. *PLA folding*: for a matrix with a 1, $\max(\text{pw}(G_M) + 1, \lceil |N|/2 \rceil) \leq \text{pla}(M)$, and the gap to $\text{pw} + 1$ is unbounded. The 5×5 identity needs 3 tracks while $\text{pw}(G_M) + 1 = 1$, and the path P_n as a matrix, one gate per edge, has $t = 2$ and $\text{pla} \geq \lceil n/2 \rceil$, so the gap is unbounded on connected instances too ([Möhring, 1990, Prop. 3.15](#)) ([Appendix A, A.13](#)).
2. *Edge separation read as cutwidth*: $\text{pw} \leq \text{cw}$ always, but $\text{cw}(K_{1,n}) \geq n/2$ and $\text{mcw}(K_{1,n}) \geq n/2 - 1$ while $\text{pw}(K_{1,n}) = 1$; so neither cutwidth reading is within a constant of pathwidth ([Appendix A, A.14](#)).

For the table, these rows are false as stated. Each has an exact variant ([Theorem 2.4](#)): the PLA claim holds for multiple folding, and the edge-separation reference proves exactly the vertex separator game.

Several published equalities need a nondegeneracy hypothesis, and one published proof has a gap that we repair; [Appendix B.2](#) gives the details. [Appendix B.3](#) adds a thirteenth problem, outside the table, to the exact core.

Summary. Eight of the twelve rows of the Linhares–Yanasse table are exact, counting the pathwidth row itself, which is the reference. The other seven are pathwidth, or pathwidth plus one, exactly: MOSP, gate matrix layout, one-dimensional logic, interval thickness, node search, narrowness and vertex separation. So are two variants of the false rows, multiple folding and Lengauer’s vertex separator game. Split bandwidth and edge search are bands of width one and two, within one of the number of open stacks but not at a fixed offset. Simple PLA folding and edge separation are false as stated. One theorem is stated and not proved, LaPaugh’s $\text{es} = \text{pes}$, and no row needs it. One of the table’s references, Kashiwabara and Fujisawa (1979), is not held; its row is proved from [Möhring \(1990\)](#) and [Kirosis and Papadimitriou \(1985\)](#).

¹ $\text{sb}(K_{1,3}) \geq 2$ is also argued by hand: splits of a tree are trees and keep at least three leaves.

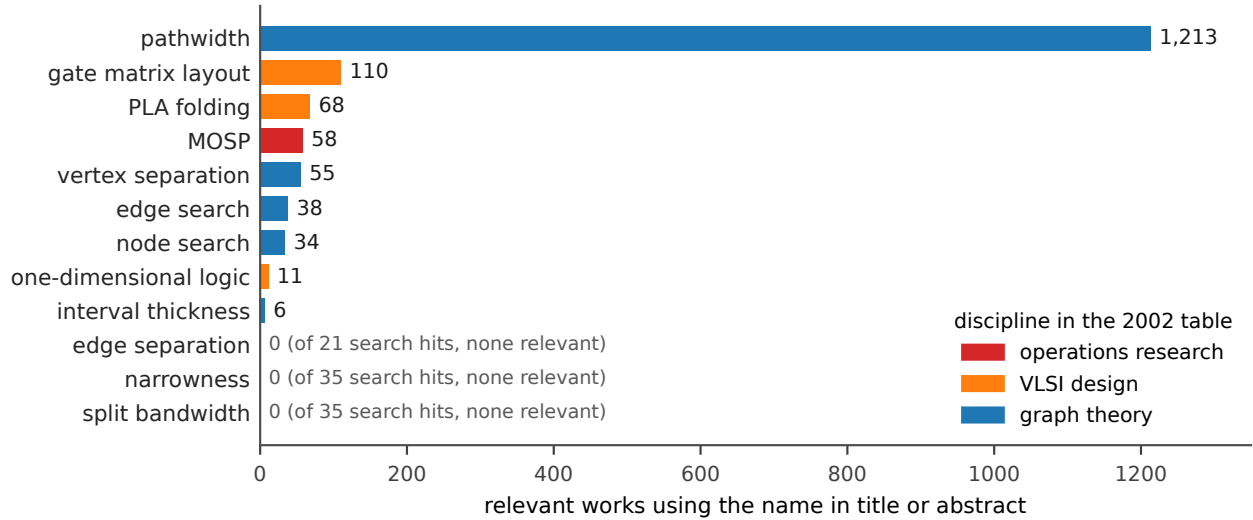


Figure 3.1: Pathwidth has more relevant works than the other eleven names together, three times over, and three names have none. Works using each problem’s name, in title or abstract, in the sense of the Linhares–Yanasse table; all years to September 2026. Colour gives the discipline that the table assigns the problem.

3 The names and the communities

Section 2 showed that eight of the twelve rows of the Linhares–Yanasse table are one quantity. A result proved under any of their names therefore holds under all of them. But a reader who searches under one name will not find a result published under another. This section asks two questions about the literature. Which name does it use for the quantity? And, besides that name, which community has been most active? The answers are pathwidth, by far, and open stacks. The second answer explains where the best exact algorithm for the family came from, and leads to Section 4.

How we counted. We searched OpenAlex (Priem et al., 2022) on 29 September 2026 for each problem’s name and its common variants, in titles and abstracts, within Computer Science, Mathematics, Engineering and Decision Sciences. A search hit need not use the name in the table’s sense: “narrowness” matches term rewriting, and “path-width” matches roads and tool paths. So we read the 2,271 hits and kept the 1,593 that are about one of the twelve problems. The 700 hits for the eleven smaller names were judged one by one, by one annotator with model assistance; the 1,571 pathwidth hits were filtered by a stated rule, checked on samples. A mechanical phrase rule agrees with the judgements on 78% of the 700 and leads to the same conclusions. The counts are lower bounds, since a work can study a problem without naming it in its abstract. Appendix C gives the method in full, the robustness check and the counts per name.

3.1 Pathwidth has absorbed the family

Figure 3.1 gives the counts. Pathwidth has 1,213 relevant works, 21 times as many as MOSP and more than three times as many as the other eleven names together (380). Three names have no relevant work at all: narrowness, split bandwidth and edge separation. Two more are rare: interval thickness has 6 works and one-dimensional logic 11. Pathwidth is also the only name still growing: its 311 works in 2020–2024 are more than the whole history of any other name (Appendix C). So anything one would want to know about algorithms for this class of problems was most likely published under “pathwidth”.

Table 3.1: Besides pathwidth, the most active name over 2005–2024 is MOSP, although vertex separation led in 2015–2019. Relevant works per five-year period, for the three names with the most works over 2005–2024; the next are edge search (28) and node search (20).

Name	2005–09	2010–14	2015–19	2020–24	2005–24
Pathwidth	114	179	267	311	871
MOSP	17	17	5	12	51
Vertex separation	7	10	15	0	32

Table 3.2: The open-stacks papers are read apart from graph theory. Works citing the papers that the Linhares–Yanasse table cites, by the discipline of the papers they cite. Of 135 works citing an operations-research paper, 6 also cite a graph-theory paper (4%); of 285 citing a VLSI paper, 65 do (23%). The 2002 paper itself counts as operations research. OpenAlex, 29 September 2026; 844 citing works in all.

Works citing	All	... of which also cite		
		operations research	VLSI	graph theory
operations research	135	–	14	6
VLSI	285	14	–	65
graph theory	507	6	65	–

3.2 Besides pathwidth, open stacks

The VLSI names belong to the 1980s: since 2005 gate matrix layout has 9 relevant works and PLA folding 6. Table 3.1 shows the three names with the most works over 2005–2024. MOSP is second, with 51, ahead of vertex separation (32) and edge search (28). Vertex separation led only in 2015–2019. The open-stacks community, in operations research, is thus the one most active besides pathwidth.

It has also worked largely apart from graph theory. Table 3.2 counts the 844 works that cite the papers of the table, by the disciplines of the papers they cite. Only six works cite both an open-stacks paper and a graph-theory paper. Two are by the table’s authors, one is open-stacks work on graph properties, and three are pathwidth or vertex-separation work from computer science. The VLSI papers are read with graph theory about five times as often.

3.3 From an open-stacks challenge to a pathwidth solver

It is not a coincidence that the best exact algorithm for MOSP came from the open-stacks community and is also, on the published evidence, among the strongest exact algorithms for pathwidth. The account is in [Chu and Stuckey \(2009, §1\)](#). MOSP was the subject of the first Constraint Modelling Challenge ([Smith and Gent, 2005](#)), posed in May 2005, which drew 13 entries. The winning entry, by García de la Banda and Stuckey, combined dynamic programming with branch and bound and was an order of magnitude faster than any other ([García de la Banda and Stuckey, 2007](#)). [Chu and Stuckey \(2009\)](#) built on it. They branched on which customer’s stack closes next rather than on which product is made next, added nogood recording and dominance rules, and became five to six orders of magnitude faster on hard instances. Their solver closed every open instance of the challenge within ten seconds. MOSP’s busiest decade in Table 3.1, 2005–2014, is the decade of this work.

The customer search reads an instance only through its MOSP graph ([Chu and Stuckey, 2009, p. 4](#)), so it decides pathwidth (Section 4). Ported to graphs, it proves more instances than reported on the benchmarks where published exact pathwidth results exist, those of [Coudert et al. \(2016\)](#). It proves the pathwidth of 11,194 of the 11,534 Rome graphs (97.1%), against their 95.6%, both within 600 seconds per graph. It

proves all 50 VSPLIB trees, against their 30, and 39 of the 73 Harwell–Boeing graphs, against their 26. These are comparisons with published numbers obtained on other hardware, not a head-to-head run.

Two consequences shape the rest of the paper. First, a result on any of the twelve problems should be stated, and searched for, as a result on pathwidth. Section 4 does this for the customer search. It finds that the search’s first dominance rule is, in correct form, a theorem already known in the pathwidth literature, and that two of its published theorems are false. Second, a benchmark for any problem in the exact core is a pathwidth benchmark, which is how Section 5.4 builds its dataset.

4 An exact open-stacks search, read as a pathwidth algorithm

If MOSP is pathwidth (Theorem 2.3), then the best exact MOSP algorithm is a pathwidth algorithm, and its rules can be checked against what the pathwidth literature knows. This section does that for the *customer search* of Chu and Stuckey (2009), the search that Section 3 traced from the open-stacks challenge. It searches over the orders in which the customers’ stacks close, not over pattern sequences, and it prunes with dominance rules. Three of the rules are stated as theorems (their Theorems 1, 2 and 3) and a fourth, the subset rule, in their §2. Chu and Stuckey (2009) place MOSP among “graph path-width and gate matrix layout”, citing Linhares and Yanasse (2002), but they do not use pathwidth in the search. Any graph is the MOSP graph of the instance with one customer per vertex and one pattern per edge, so the search decides pathwidth for every graph.

The stakes are these. Two of the published dominance theorems, the *definite move* and the *better move*, are false, and so are their restatements in Chu’s thesis (Chu, 2011) and, with a stronger premise, in Fink (2012). Yet no wrong answer is known: no whole instance on which the published rules answer “no” wrongly has been found, and when every benchmark optimum that rested on the search was proved again with repaired rules, no value changed (Section 5). The correct form of the definite move was already known for pathwidth, as Tamaki’s commitment lemma, and we show that it can be tested by one small bipartite matching per candidate. The main result, Theorem 4.8, says informally: *with both rules repaired, the search is sound; whenever it answers that k stacks do not suffice, they do not, and the graph has pathwidth at least k* . The repair costs at most 2% more search nodes on each benchmark set (Section 5).

Section 4.1 defines the search and shows that it decides pathwidth, in both vocabularies. Section 4.2 gives the algorithm. Sections 4.3 and 4.4 take the rules one at a time, Section 4.5 proves the repaired search sound, and Section 4.6 places each rule in the pathwidth literature.

4.1 The search decides pathwidth

Chu and Stuckey (2009, p. 4) observe that the patterns matter only through the MOSP graph. So fix a graph $G = (C, E)$ whose vertices are the customers, and an integer $k \geq 0$. In this section k bounds the number of open stacks, which is one more than a width: k stacks suffice exactly when the pathwidth is at most $k - 1$ (Theorem 4.2). For a customer c write $N[c]$ for its closed neighbourhood, the stacks that must be open when c ’s stack closes. For a set $T \subseteq C$ of vertices, its *border* ∂T is the set of vertices outside T with a neighbour in T . For a set T of closed customers write

$O(T) = \bigcup_{c \in T} N[c]$	the stacks opened once T has closed,
$\text{fn}(X) = \{c : N[c] \subseteq X\}$	the customers free to close when X is open,
$\text{cl}(T) = \text{fn}(O(T))$	the <i>free closure</i> of T ,
$b(T) = O(T) \setminus T = \partial T $	the stacks open after T has closed,
$o(c, T) = N[c] \setminus O(T), \quad \text{open}(c, T) = o(c, T) $	the stacks closing c would newly open.

Table 4.1: The customer search is, ingredient by ingredient, an exact pathwidth algorithm: each notion of the search beside its name in the pathwidth literature. The memo and the definite move are described in Section 4.2.

Customer search	Layout
closing order from $\emptyset$	vertex sequence, read forwards
open stacks $b(T)$	border size $ \partial T $
$\text{cost}(T, c)$	$ \partial(T \cup \{c\}) + 1$
$\text{Sol}_k(\emptyset)$	$\text{vs}(G) \leq k - 1$
free closure $\text{cl}(T)$	full set of T (Suchan and Villanger, 2009, p. 328)
memo	failure table
definite move, repaired	commitment to the child

The open stacks after T has closed are exactly the border of T , and this one identity carries the whole translation to pathwidth.

A *move* at T closes one customer $c \notin T$, at cost $\text{cost}(T, c) = |O(T \cup \{c\}) \setminus T| = b(T \cup \{c\}) + 1$, the stacks open at that moment counting c . It is *playable* when $\text{cost}(T, c) \leq k$. Write $P_k(T)$ when some order of $C \setminus T$ has every step playable. On entering a node the search closes every free customer, so it stands only at *states*, sets with $S = \text{cl}(S)$. The *child* of S by c is $S \cdot c = \text{cl}(S \cup \{c\})$. The search's predicate $\text{Sol}_k(S)$ holds when $S = C$, or when some playable c has $\text{Sol}_k(S \cdot c)$ (Appendix A, A.15). Every node the search visits satisfies the invariant $b(S) \leq k$.

An example. Take the graph of Figure 1.1(d), with $k = 3$. At $T = \{a\}$ we have $O(T) = \{a, b\}$, so one stack, b , is open: $b(T) = 1$ and $\partial T = \{b\}$. No other customer is free, since $N[b]$ contains c and d , so T is a state. Closing b newly opens $o(b, T) = \{c, d\}$ and costs $\text{cost}(T, b) = 3$, the stacks b, c and d ; the move is playable. At $\{a, b, c\}$ every neighbour of d is open, so d is free, and the child $\{a, b, c\} \cdot c$ is $\{a, b, c, d\}$. The closing order a, b, c, e (with d and f closed free) costs 2, 3, 3, 2, the profile of order A in Figure 1.1(a).

Lemma 4.1 (free moves never hurt). *If $b(T) \leq k$, then $P_k(T) \iff \text{Sol}_k(\text{cl}(T))$ (Appendix A, A.16).*

Closing a customer whose neighbourhood is already open opens nothing, never makes a later step dearer, and leaves one fewer stack open. The hypothesis is needed: closing the free customers one by one costs $b(T)$ at each step.

Theorem 4.2 (the search decides pathwidth). *For nonempty C , $\text{Sol}_k(\emptyset) \iff \text{pw}(G) + 1 \leq k$. On the MOSP graph of an instance M with at least one 1, $\text{Sol}_k(\emptyset) \iff Z(M) \leq k$ (Appendix A, A.17).*

Proof. For a closing order $c_1, \dots, c_n$ with prefixes S_i , the i -th step costs $1 + |\partial S_i|$. Reverse the order to get a layout. Its first $n - i$ vertices are $C \setminus S_i$, and those with a neighbour among the remaining ones are exactly ∂S_i . So the cost of the closing order is the vertex separation of the reversed layout plus one, and Lemma 4.1 at the root and Theorem 2.2 finish the proof. $\square$

So a refutation, $\neg \text{Sol}_k(\emptyset)$, may only mean $\text{pw}(G) > k - 1$. The rest of the section asks whether the pruning rules preserve that meaning.

In pathwidth language. The exact pathwidth literature writes $N(T)$ for the border and $d(T)$ for its size (Kobayashi et al., 2014, p. 390); we keep ∂T and $b(T)$. A vertex sequence is *w-feasible* when every prefix has border at most w , and $\text{vs}(G)$ is the least w for which some permutation is w -feasible. Table 4.1 translates each notion of the search into this vocabulary. Read it row by row: every ingredient of the customer search is a known ingredient of an exact pathwidth algorithm, and the last two rows are the rules of Section 4.2.

Algorithm 4.1 The customer search (Chu and Stuckey, 2009). It decides $\text{Sol}_k(S)$ for a graph $G = (C, E)$ and a stack bound k , and is called as $\text{SEARCH}(\emptyset, \emptyset)$. Q is the set of old moves the node inherits.

```

1  SEARCH( $S, Q$ ):
2     $S \leftarrow \text{cl}(S)$                                 free moves (Lemma 4.1)
3    if  $S = C$  then return true
4    if  $S$  is in the memo then return false            memo
5     $P \leftarrow$  the  $c \notin S \cup Q$  with  $\text{cost}(S, c) \leq k$ , in index order    cost cut
6     $L \leftarrow$  filter of  $P$ : definite move, then subset rule, then better move    Sections 4.3–4.4
7     $\text{seen} \leftarrow Q \setminus S$ 
8    for each  $c \in L$ :
9       $Q' \leftarrow \{q \in \text{seen} : \text{cost}(S \cup \{q\}, c) \leq k\}$         old moves
10     if  $\text{SEARCH}(S \cup \{c\}, Q')$  then return true
11      $\text{seen} \leftarrow \text{seen} \cup \{c\}$ 
12     add  $S$  to the memo
13   return false

```

4.2 The algorithm

Algorithm 4.1 gives the customer search. It is a depth-first search over states. At each node it closes the free customers, consults a table of states already refuted, keeps the playable candidates, filters them with the dominance rules and recurses on the survivors.

Line 2 is safe by Lemma 4.1, and the cost cut of line 5 is exact, since an unplayable move is not a move of Sol_k . The rest is pruning, and each piece has a one-sentence intuition.

- *The memo* (lines 4 and 12): whether a state can be finished does not depend on how it was reached, so a state refuted once is refused when met again.
- *Old moves* (lines 7–11): a sibling whose subtree failed stays failed further down, as long as closing it first would have kept each later step within the bound.
- *The definite move*: if closing q now leaves no more stacks open than before, it seems safe to close q at once and try nothing else.
- *The subset rule*: if d newly opens a subset of what r newly opens, then d is at least as good a move as r .
- *The better move*: if q would be a definite move right after r , then closing q before r is at least as good as closing r first.

Each of the three filtering rules discards a move r by citing a *covering*, a kept move c with $\text{Sol}_k(S \cdot r) \Rightarrow \text{Sol}_k(S \cdot c)$. The filter runs the definite move, then the subset rule, then the better move, each once. If the definite move fires, it keeps its customer alone.

This loop is the decision procedure of Kobayashi et al. (2014, Algorithm 1, p. 393), which asks whether a vertex set extends to a w -feasible permutation: replace the set by a committed extension, answer from a failure table or at the full vertex set, recurse on each feasible one-vertex extension, and record failures. The customer search is that loop with $w = k - 1$. The free closure and the definite move play the part of the commitment, the memo is the failure table, and the cost cut is the feasibility test. Coudert et al. (2014, §3.2) has the same three parts in a branch and bound. Neither line cites the other in the papers we hold: Chu and Stuckey (2009) predates the commitments and both 2014 papers, and none of the pathwidth papers cites it. This is the isolation measured in Table 3.2, at the level of one algorithm.

A counterexample to Chu & Stuckey's Theorem 1, at $k = 6$
0 is playable and $\text{close}(0, S) \geq \text{open}(0, S)$, so the rule keeps 0 alone. A solution from S exists
(first moves 1, 3, 4, 6), but none starts with 0.

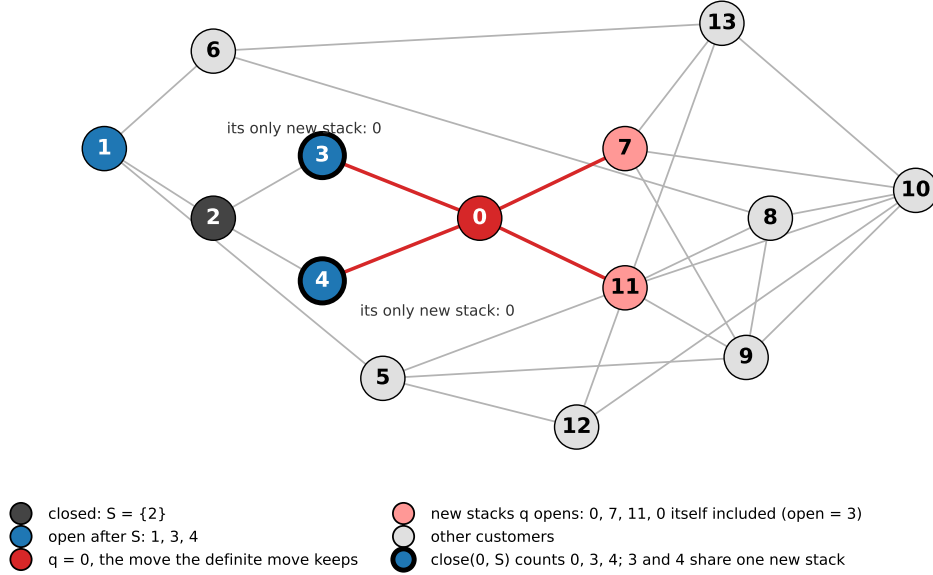


Figure 4.1: The graph of Counterexample 4.4 at the state where the definite move fails. Customer 2 is closed; 1, 3 and 4 are open. Moving 0 opens the new stacks 0, 7 and 11 and closes 0, 3 and 4, so $\text{open} = \text{close} = 3$ and the rule keeps 0 alone. But 3 and 4 have the single new stack 0 between them: closing them first gains two stacks for the price of one, which a solution starting with 0 cannot match.

4.3 The definite move is false as published, and its repair is a matching

The definite move is the strongest of the rules, since when it fires it discards every other candidate. Its premise looks like a safe accounting: closing q opens some new stacks and lets some customers close, and if it lets at least as many close as it opens, nothing seems lost by closing it first. Chu and Stuckey (2009, §3.1, p. 6) state:

Theorem 1. Suppose $S ++ [q]$ is playable and $\text{close}(q, S) \geq \text{open}(q, S)$, then if $U' = S ++ R$ is a solution, there exists a solution $U = S ++ [q] ++ R'$.

Here $++$ is concatenation of closing sequences, a *solution* is a complete closing order with every step playable, and $\text{close}(q, S)$ counts the customers d with $o(d, S) \subseteq o(q, S)$ (p. 4). The rule keeps q alone. We read d as ranging over customers not yet closed, as our implementation does; read literally, closed customers count too, which weakens the premise, and both readings are refuted below. Chu (2011, Thm 6.3.6, p. 142) restates the theorem with the same premise and proof, reworded for k stacks. Write $X = S \cdot q$. The first lemma confirms that the premise says what the intuition says.

Lemma 4.3 (what the premise says). $\text{close}(q, S) = |X \setminus S|$. Hence the premise holds iff $b(X) \leq b(S)$: the child has no more open stacks than the parent (Appendix A, A.24).

The premise compares only the two ends, S and X . The rule fails because a set between them can be cheaper still. Figure 4.1 shows the graph on which it fails.

Counterexample 4.4. Let G be the graph of Figure 4.1, on customers $0, \dots, 13$ with the 26 edges

0–3, 0–4, 0–7, 0–11, 1–2, 1–5, 1–6, 2–3, 2–4, 5–9, 5–11, 5–12, 6–8,
6–13, 7–9, 7–10, 7–13, 8–9, 8–10, 8–11, 9–10, 9–11, 10–11, 10–12, 10–13, 12–13,

with $S = \{2\}$, $k = 6$ and $q = 0$. Then $O(S) = \{1, 2, 3, 4\}$, $o(0, S) = \{0, 7, 11\}$ and $\text{cost}(S, 0) = 6$, so 0 is playable. The customers with $o(d, S) \subseteq \{0, 7, 11\}$ are 0, 3 and 4, so $\text{close}(0, S) = 3 \geq \text{open}(0, S)$, and 0 comes first in index order, so the rule keeps only 0. A solution exists from S : the order

$$1, 3, 4, 6, 12, 13, 0, 5, 7, 8, 9, 10, 11$$

costs at most 6. None exists from the child $X = \{0, 2, 3, 4\}$: only seven sets are reachable from X by moves of cost at most 6, and none is C (Appendix A, A.25).

The published implication, as a universal statement, is refuted in Lean under both readings of close, and so is the thesis's Theorem 6.3.6 (Appendix A, A.26). Fink (2012, pp. 27–28) restates the rule with a stronger premise, counting only dominated customers of smaller index than q . The graph above refutes it under no labelling, but a third twin of 3 and 4 gives a 15-customer graph that does (Appendix A, A.27). The published proof moves q to the front of a solution and counts, after each prefix, the stacks q opens against the customers it lets close. But customers that the solution has already closed no longer count as closed by q , while the stacks q opens may still be new. Appendix B.4 gives the full analysis, and Appendix B.5 the search for smaller counterexamples.

The repair. The fix is to require the comparison at every set between the two ends, not only at S . Call q *hereditarily definite* at S when $b(X) \leq b(B)$ for every B with $S \subseteq B \subseteq X$ and $q \notin B$. By Lemma 4.3, the published premise is the case $B = S$. In the counterexample the repair fails at $B = \{2, 3, 4\}$, where $b(B) = 2 < 3 = b(X)$ (Appendix A, A.28). The proof that the repair is sound needs one inequality.

Lemma 4.5 (open stacks are submodular). *For all $A, B \subseteq C$, $b(A \cup B) + b(A \cap B) \leq b(A) + b(B)$ (Appendix A, A.29).*

Proof. $b(T) = |O(T)| - |T|$, $O(A \cup B) = O(A) \cup O(B)$ and $O(A \cap B) \subseteq O(A) \cap O(B)$; the customers themselves balance exactly. $\square$

Theorem 4.6 (the repaired definite move is sound). *If $q \notin S$ is hereditarily definite at S , then $P_k(S) \Rightarrow P_k(X)$. At a state with the invariant, $\text{Sol}_k(S) \Rightarrow \text{Sol}_k(S \cdot q)$ (Appendix A, A.30).*

Proof. Take an order from S of cost at most k with prefixes T_i , in which q is the j -th step, and follow it from X , skipping customers already in X . A step $i < j$ closes c_i at $T_{i-1} \cup X$. We have $S \subseteq T_i \cap X \subseteq X$ and $q \notin T_i$, so the hypothesis gives $b(X) \leq b(T_i \cap X)$, and Lemma 4.5 gives $b(T_i \cup X) \leq b(T_i)$: the step costs no more than the original. After step $j - 1$ the closed set contains T_j with the same opened set, and the rest of the order applies. The search's form follows by Lemma 4.1. $\square$

This theorem is not new. The hereditary premise is the commitment condition of Tamaki, and the proof is the proof of the commitment lemma, Kitsunai et al. (2016, Lemma 1, p. 142); Section 4.6 makes the correspondence exact.

The matching test. The repaired rule seems to require checking exponentially many intermediate sets. It does not, and this cheap test is the algorithmic contribution of the section; we did not find it in the papers we hold. In words: the rule is safe exactly when all but one of the new stacks can be paired with a different freed customer who needed it. In the counterexample, closing 0 opens three new stacks and frees 3 and 4, both of whom needed only stack 0: one pairing where two are required.

Theorem 4.7 (the repair is a matching condition). *Let $Y = X \setminus (S \cup \{q\})$, and join each $d \in Y$ to the stacks of $o(d, S) \subseteq o(q, S)$. Then q is hereditarily definite at S iff this bipartite graph has a matching with at least $\text{open}(q, S) - 1$ edges (Appendix A, A.31).*

Proof. The sets of the repair are $B = S \cup D$, $D \subseteq Y$, and $b(X) \leq b(S \cup D)$ iff $|D| - |o(D)| \leq \delta$ with $\delta = |Y| + 1 - \text{open}(q, S)$, where $o(D)$ is the union of the $o(d, S)$. By the deficiency form of Hall's theorem the largest matching has size $|Y| - \max_D(|D| - |o(D)|)$, which is at least $|Y| - \delta = \text{open}(q, S) - 1$ exactly when the repair holds. $\square$

The bipartite graph has at most $\text{close} - 1$ customers on one side and open stacks on the other, and the matching can stop at $\text{open} - 1$ edges. Every q with $\text{open}(q, S) \leq 1$ passes at once, so the published rule is right there (Appendix A, A.32).

4.4 The other rules

Each of the remaining rules is quoted, proved sound in the form used, and shown to need the test or tie-break it carries.

The subset rule. “If $o(c_i, S) \subseteq o(c_j, S)$ and $i < j \dots$ move j can be removed” (Chu and Stuckey, 2009, §2, p. 5). As implemented, d dominates r when $o(d, S) \subseteq o(r, S)$ and the inclusion is strict or $d < r$. Every dominated playable r is dropped, with a fallback that keeps P if all would go. The covering is sound for every set S : if $o(d, S) \subseteq o(r, S)$ and r is playable, then d is playable and $\text{Sol}_k(S \cdot r) \Rightarrow \text{Sol}_k(S \cdot d)$, because d is free after r and closing d then r reaches the same state at no greater cost. With the index tie-break, domination is a strict order, so an undominated move with a solution survives, and the fallback never fires when a solution exists (Appendix A, A.33). The tie-break is necessary: without it, two customers with equal new stacks dominate each other and both go, which loses a solution on a 7-customer graph (Appendix A, A.34).

The better move. Chu and Stuckey (2009, §3.2, p. 6): “Suppose $S \mathrel{++} [q]$ and $S \mathrel{++} [r, q]$ are playable and $\text{close}(q, S \cup \{r\}) \geq \text{open}(q, S \cup \{r\})$ then if $U' = S \mathrel{++} [r] \mathrel{++} R$ is a solution there exists a solution $U = S \mathrel{++} [q] \mathrel{++} R'$.” Its premise is the definite move's premise for q at the child $\text{cl}(S \cup \{r\})$, together with playability of r then q , and its conclusion swaps q and r (Appendix A, A.35). The swap is sound, because both orders end at the same state and neither step overflows. The appeal to the definite move is not.

Counterexample. On the graph of Counterexample 4.4 at the root with $k = 6$, the definite move does not fire, customers 0 and 2 both survive the subset rule, and the premise holds for $r = 2$, $q = 0$, so the rule drops 2 citing 0. But $\{2\}$ has a solution and $\{0\}$ has none (Appendix A, A.36). At that node another kept customer still has a solution, so this is a false link and not a lost node; no node where the better move loses the last solution has been found. Chu (2011, Thm 6.3.8, p. 143) states the better move with a different premise, $\text{close}(q, S) \geq \text{open}(q, S \cup \{r\})$, and proves it through his Theorem 6.3.6. The same graph refutes it at $S = \{2\}$, $r = 3$, $q = 0$ (Appendix A, A.37).

Repair and test. Cite q for r only if q is hereditarily definite at $\text{cl}(S \cup \{r\})$, which is one matching at the child (Theorem 4.7); then the swap argument goes through (Appendix A, A.38).

Two requirements. The close count in the premise must not include the customers that r finishes on its own; counting them makes the rule's own conclusion false on a 12-customer graph. And the rules must run in the order definite, subset, better, each citing only customers still standing: if the better move runs first, the subset rule can cite a customer the better move has already dropped, and an 8-customer graph loses a node through a cycle of links that are each true (Appendix A, A.39). The second is an instance of the general observation of Beck et al. (2025, Prop. 3, Example 6) that mutually dominating transitions must be tie-broken.

The old move. Chu and Stuckey (2009, Thm 3, p. 7) let a child inherit a sibling q whose subtree failed, provided that inserting q before each later step keeps that step playable (line 9 of Algorithm 4.1). The rule is sound as implemented: if $\text{cost}(S \cup \{q\}, c) \leq k$, then $\text{Sol}_k(\text{cl}(\{q\} \cup S \cdot c)) \Rightarrow \text{Sol}_k(S \cdot q)$, so a refuted sibling stays refuted along any path whose steps pass the test (Appendix A, A.40). The test is necessary; the smallest graph that shows it is the path on five vertices (Appendix A, A.41).

The memo. Whether a state with $b(S) \leq k$ has a solution depends on the state alone, not on the path that reached it (Appendix A, A.42). One concern is that the memo and the old move might not combine: a refutation obtained with inherited old moves seems to depend on the ancestors, and one implementation of the search declines to run both for that reason. A node taken on its own, with an old move that is not in fact refuted, can indeed record a solvable state (Appendix A, A.43). Theorem B.2 (Appendix B.7) shows that this never happens in a real run, because every old move is a refutation completed earlier in the same run.

4.5 The repaired search is sound

What a rule must satisfy. At a state S with old moves Q , the cost cut keeps the playable candidates P and the filter keeps a sublist $L \subseteq P$. The filter is *node-sound* at (S, Q) if, whenever every $q \in Q$ has $\neg \text{Sol}_k(S \cdot q)$, $\text{Sol}_k(S)$ implies $\text{Sol}_k(S \cdot c)$ for some $c \in L$ (Appendix A, A.44). In terms of coverings, a node is sound when two things hold. First, every link a rule cites is a true covering. Second, every chain of links from a discarded candidate ends in L or in Q . The second condition can fail through a cycle even when every link is true, which is why the order of the rules matters. Proposition B.1 shows that the repaired filter is node-sound, and Theorem B.2 that node soundness, the memo and the old move together give sound runs; both are in Appendix B.7. Together they give the main theorem.

Theorem 4.8 (the revised search is sound). *Run the customer search with free moves, the memo, old moves and the filter definite $\rightarrow$ subset $\rightarrow$ better, the definite move firing only on a hereditarily definite customer and the better move citing only under the repaired premise. If it refutes k from the root, then no closing order costs at most k , $\text{pw}(G) > k - 1$, and on the MOSP graph of an instance M with at least one 1 , $Z(M) > k$ (Appendix A, A.45).*

In words: each rule throws away only moves that a kept move is proved to be at least as good as, and the memo and the old moves remember only failures that really happened earlier in the same search. So when the search says “no”, no closing order fits in k stacks.

What is proved and what is tested. The theorem is about the search as specified; that our C and Python implementations run it is checked, not proved. Three checks link them:

- a transcription of the Lean filter agreed with the implemented filter on 16,244,090 node checks over graphs on 1 to 17 vertices;
- the matching agreed with the hereditary premise on 36,939,226 candidates;
- a search written from the theorems agreed with both implementations and an exact oracle on 17,431,232 runs over 52,592 graphs on 1 to 17 vertices.

Section 5.3 adds a certificate checker for individual refutations.

The published rules. Under the published rules the filter is not node-sound (Appendix A, A.46). In a check of 570,206 runs with memo and old move, it lost the last solution at a node in 58 runs, all on graphs built around Counterexample 4.4 and all at k equal to the optimum. Every one found a solution through another branch and answered correctly. No whole instance on which the published rules answer *unsat* wrongly is known.

4.6 The rules in pathwidth language

The definite move has an exact counterpart in the pathwidth literature, and this subsection makes the correspondence precise. Kitsunai et al. (2016, p. 142), stating Tamaki’s notion, call an extension τ of a vertex sequence σ *committable* when $|\partial X'| \geq |\partial V(\tau)|$ for every X' with $V(\sigma) \subseteq X' \subseteq V(\tau)$. Their Lemma 1 says that if σ extends to a feasible permutation, so does τ .

Lemma 4.9. *For $q \notin S$ and $X = \text{cl}(S \cup \{q\})$: the published premise holds iff $b(X) \leq b(S)$, the commitment condition at the single set $X' = S$; and the repaired premise holds iff X is a commitment from S , the condition at every X' between them (Appendix A, A.47).*

So Theorem 4.6 is the commitment lemma at the child, and the published definite move is a commitment whose interior condition was dropped. In Counterexample 4.4 the endpoint test passes with $b(X) = 3 = b(S)$, while the interior set $\{2, 3, 4\}$ has border 2 (Appendix A, A.48). The published premise is not wasted: by Kitsunai et al. (2016, Lemma 10, Cor. 2, pp. 148–149) it guarantees a commitment to the least-border set between S and X , which need not contain q (Appendix A, A.49). In the counterexample that set is $\{2, 3, 4\}$ (Appendix A, A.50). Committing there when the matching fails is a second repair that gives up no pruning; we have not built it.

A commitment of depth one, S to $S \cup \{q\}$, holds iff $\text{open}(q, S) \leq 1$ (Appendix A, A.51). The greedy step of Coudert et al. (2014, Lemma 3, p. 49) is exactly that condition (Appendix A, A.52), and the push rule of Suchan and Villanger (2009, Rule 1, p. 330) is a case of it. So every published pathwidth rule of this kind lies in the region $\text{open} \leq 1$, where the published definite move is right; Kobayashi et al. (2014, p. 388, Table 2) report depth-one commitments as “extremely effective”. Chu and Stuckey’s definite move reaches into $\text{open} \geq 2$, and that is where it fails. For the cost of the test, Kitsunai et al. (2016, Cor. 2) find the least border between the ends of a given extension by a minimum s – t separator; Theorem 4.7 is that computation for the single target X , where it becomes a bipartite matching.

Table 4.2 sums up, and its rows are grouped by what they say. The published definite and better moves are false. The free move, the memo and the repaired definite move are known pathwidth results. The subset rule, the repaired better move and the old move have no counterpart we found, and are proved here, as is the matching test. The comparison is limited to the pathwidth papers we hold (Suchan and Villanger, 2009; Bodlaender et al., 2012; Coudert et al., 2014, 2016; Kobayashi et al., 2014; Kitsunai et al., 2016); “none found” is not a claim of priority.

Prior reports. We found no earlier report of the error in the 56 works we read in full of the 141 that cite the customer search’s papers (Appendix B.6). The pathwidth literature has the correct form of the rule, as a commitment, and none of the pathwidth papers we hold mentions MOSP.

Table 4.2: Two published rules are false; the free move, the memo and the repaired definite move are known pathwidth results; the subset rule, the repaired better move and the old move, like the matching test, are proved here. Each rule of the customer search against the exact-pathwidth literature.

Rule	Published counterpart	Status
definite move, published	–	false (Cex. 4.4)
better move, published	none found	false
free move	full sets (Suchan and Villanger, 2009); Kitsunai et al. (2016, Prop. 3)	known
definite move, open ≤ 1	depth-one commitments; greedy step; push rule	known
definite move, repaired	commitment lemma (Kitsunai et al., 2016, Lemma 1)	known; matching test (Thm 4.7) not found
memo	subset DP (Bodlaender et al., 2012); prefix and failure tables	known; with the old move, proved here (Thm B.2)
subset rule	none found	proved here
better move, repaired	none found	proved here
old move	none found	proved here

5 Computational results and a dataset

Section 4 proved the repaired search sound. This section asks four practical questions about it and answers each with a measurement. What does the repair cost? Under the repaired rules, the total number of search nodes over each benchmark set changes by at most 2%. Do the published benchmark optima survive? All 115 that rested on the published rules were proved again, and none changed. Can a reader check a refutation without trusting our code? Yes, up to about 75 customers, through a certificate and an independent checker. Finally, what does it make possible? A dataset of 17,714 graphs from twenty benchmark collections. For 16,087 of them the pathwidth is certified, and that gives the exact value of every problem in the exact core.

5.1 Experimental setup

Machine and code. Every run in this section used one machine, an AMD Ryzen 9 9950X (16 cores, 32 hardware threads) with 128 GB of memory under Ubuntu Linux. Both searches, the open-stacks customer search and its port to graphs, are written in C (gcc 15.2, `-O3 -march=native`) and driven from Python 3.12. A Python implementation of the same rules serves as a reference for cross-checks and emits the certificates of Section 5.3. The repaired tests are the default in every implementation; the published tests remain available as an option.

Runs. The basic job asks one question: is the optimum of this instance at most k ? A *worker* is one single-threaded process that answers such questions, one at a time. A *node* is one state visited by the search, the unit in which we report effort. Up to 32 workers ran at once on the machine, so wall-clock times carry load noise. We therefore compare node counts and report time only in core-hours. A *pair* runs the same question twice in the same worker, once under the published rules and once under the repaired rules, alternating which goes first. A *descent* solves a whole graph: it lowers k from an upper bound until a refutation proves the last value optimal. Each table states its time or node budget.

Table 5.1: The repair never changed an answer and changes the total node count by at most 2%. Each pair runs the same question under the published and the repaired rules in one worker. Totals are over the pairs that finished under both rules; a pair stopped by its budget says nothing about cost. *Largest pair ratio*: the largest ratio of repaired to published nodes over single pairs. Open-stacks pairs refute the optimum minus one, at 600 s per call at 50–100 customers and 2×10^8 nodes at 125. The graph pairs run whole descents on VSPLIB, the colouring and named graphs, and a sample of 500 Rome graphs.

Instances	Vertices	Pairs finished	Nodes, published	Nodes, repaired	Total ratio	Largest pair ratio
<i>Open stacks: one refutation of the optimum minus one per pair</i>						
MOSP corpus, base	9–40	6,135 of 6,135	228,147	228,154	1.00003	1.009
MOSP corpus, with better move	9–40	6,135 of 6,135	202,962	202,971	1.00004	1.040
Chu & Stuckey, with better move	50–100	118 of 125	1,358,096,671	1,362,205,210	1.00303	1.013
Chu & Stuckey, cap 2×10^8 nodes	125	11 of 23	90,969,508	90,961,419	0.99991	1.000
<i>Graph pathwidth: one whole descent per pair, 120 s each</i>						
VSPLIB, colouring, named, 500 Rome	4–2,916	731 of 880	2,977,849,807	3,038,670,717	1.020	–

Configurations. The open-stacks runs use two configurations of the rules of Section 4, which differ only in the better move. The *base* configuration leaves it off. The configuration *with better move* switches it on for instances with at most five products per customer on average. This is the setting under which the customer search certified the benchmark values it proved.

Instances. The *MOSP corpus* is the 6,376 instance files of five public open-stacks collections: the 2005 Constraint Modelling Challenge (Smith and Gent, 2005) with a second copy of 46 of its files, the random instances of Chu and Stuckey (2009) (30 to 125 customers), those of Faggioli and Bentivoglio (1998), and the 24 industrial SCOOP instances. They are the first rows of Table 5.3. A Chu–Stuckey name such as Random–125–125–2–4_0 gives the number of customers, the number of products, the generator’s density class and the instance number. The graph benchmarks are those of Section 3 and Table 5.3.

Provenance. Every stored value comes with a witness, an order that attains it, and one of three provenances:

- *certified by search*: a search proved that no order does better, by refuting the value minus one;
- *certified by bound*: a lower bound equals the witness’s value, so no search was needed;
- *upper bound only*: the witness is all there is, and optimality is not proved.

Of the 6,376 corpus values, 6,374 are certified and two are upper bounds only.

5.2 The cost of the repair, and the benchmark values

The repaired search adds one matching test to the definite move and to the better move (Section 4.3). A candidate that fails the matching is passed over, so the repaired tree can only differ where the matching fails. Table 5.1 shows how rarely that matters.

In every row of Table 5.1 the repaired and the published rules gave the same answer, and the total node counts differ by at most 2%. Single pairs differ by at most 4%. The cost per node is also small: about 2% in the C implementation, measured on six hard instances at 75 to 125 customers. The published test passes and the matching then fails on only 0.07–0.27% of definite-move candidates, a rate that rises with size.

The benchmark values survive. No published optimum changed under the repaired rules. Of the 6,374 certified values in the MOSP corpus, 6,259 do not depend on the customer search at all. They were proved by an exhaustive computation over all subsets of customers (up to 15 customers), by a SAT refutation whose proof an independent checker accepted (in the DRAT format, the standard proof format for SAT solvers), by an earlier SAT search, or by a lower bound equal to the value. The other 115 rested on the customer search under the published rules. The repaired search refuted the value minus one on all 115, so all 115 values stand (as of 6 October 2026).

The seven hardest are 125-customer Chu–Stuckey instances at density classes 2 and 4. They are too large for one worker, so their search was split: the top levels of the tree are expanded into independent subtrees, each refuted on its own worker, and the value is refuted when every subtree is. The split is proved sound (Appendix A, A.53). All seven were refuted at 18 to 369 core-hours each, 1,112.6 in total (Table B.1, Appendix B.8).

Two values in flight. The corpus also holds two values that were never certified by any method: 25 for Random-125-125-2-2_0 and 21 for Random-125-125-2-3_0. The same split is refuting the value minus one on each. As of 6 October 2026 it has refuted 2,141 and 2,354 subtrees, using 431 and 434 core-hours, and found no order of the smaller value. If the run finishes it certifies both values; an order of the smaller value would improve the stored value instead.

5.3 A checkable proof of a refutation

A refutation by the search is a claim about a tree of up to 10^{12} nodes, and Theorem 4.8 proves the rules sound, not the code that runs them. So the search emits a *certificate*: the tree it explored, every pruning step with its rule and its witness, and for each definite and better step the matching of Theorem 4.7. An independent checker, which shares no code with the search, reads the instance matrix and the certificate. At every node it checks that

- the premise of every pruning step holds, recomputed from the matrix;
- every matching is valid: distinct customers, distinct stacks, each stack newly opened by its customer, and at least open -1 edges;
- every candidate within the bound is either a child or covered by a pruning step, and every chain of covers ends at a child or an old move, with no cycle;

and it replays the tree from the root. Each accepted step is a Lean theorem. That an acyclic covering by such steps is sound is a short paper argument, not yet a formal proof.

Table 5.2 gives the results. Every refutation emitted within its budget verifies, 6,275 of 6,286, and none is rejected. All 35.5 MB of certificates check in 147 s. The base configuration gives the same picture: 6,276 of 6,286 verify and none is rejected. The checker also catches the error of Section 4.3. On the graph of Counterexample 4.4, rooted at $\{2\}$, the published rules emit a complete tree that claims there is no solution. A checker of the published premises accepts it. Ours rejects it, even with a largest matching attached, because the matching has one edge where two are needed. At 125 customers the trees are too large to ship as certificates.

5.4 A dataset with certified pathwidth

Section 2 showed that one certified pathwidth w answers every problem in the exact core: pathwidth, vertex separation and the vertex separator game take the value w , and MOSP, gate matrix layout with multiple folding, one-dimensional logic, interval thickness, narrowness and node search take $w + 1$. We therefore

Table 5.2: Every certificate the search emitted verifies, none is rejected, and checking takes milliseconds. Certificates for the repaired rules, configuration with better move, refuting the optimum minus one on the certified MOSP corpus at 9–75 customers, with 60 s to emit each. *Not emitted*: the emitter did not finish in time. Size is the compressed certificate; check time is the independent checker’s.

Customers	Instances	Verified	Rejected	Not emitted	Median size (bytes, gzip)	Median check (ms)
9–10	1,614	1,614	0	0	332	0.04
11–20	2,508	2,508	0	0	356	0.08
21–30	1,816	1,816	0	0	414	0.24
31–40	197	197	0	0	650	1.27
41–50	120	120	0	0	12,045	33.24
51–60	1	1	0	0	126,508	278.86
61–75	30	19	0	11	26,516	143.70

built one dataset from every benchmark collection of those problems that we could obtain, including the MOSP corpus.

The unit of the dataset is the *isomorphism class*: the set of instances whose graphs are the same up to renaming the vertices. For a matrix instance the graph is its MOSP graph. Two instances in one class have the same value under every problem, so the dataset holds one record per class, with its members and the renaming that maps each onto the record’s graph. Classes were found with nauty, a standard tool that computes a canonical form for each graph. Each record carries the graph, the width w , its provenance, and a witness layout of exactly that width. A record certified by bound also carries a *minor certificate*: a list of vertex deletions and edge contractions after which every vertex has degree at least w . Any checker can replay it, and it proves the pathwidth is at least w .

Table 5.3 gives the counts. The collections are the MOSP collections (Smith and Gent, 2005; Faggioli and Bentivoglio, 1998; Chu and Stuckey, 2009; de Carvalho and Soma, 2015; Frinhani et al., 2018), eleven VLSI gate matrix circuits (de Oliveira and Lorena, 2002), the VSPLIB sets (Duarte et al., 2012) and the Small graphs (Martí et al., 2008), the Rome graphs (Di Battista et al., 1997), and treewidth collections used as pathwidth benchmarks (van den Broek and Bodlaender, n.d.; Larisch and Salfelder, n.d.; Dell et al., 2017, 2018). Of the 17,714 classes, 16,087 carry a certified pathwidth with a witness layout and 862 a witnessed upper bound. The 610 large instances of Frinhani et al. (2018), with 400 to 1,000 customers, are included so that the dataset covers every published MOSP collection we hold; none of them has a value yet, because an exact search at that size is out of reach.

An independent checker, sharing no code with the solvers, verifies the dataset. It re-reads all 20,987 member files with its own readers, maps each onto its record’s graph by its renaming, recomputes the width of every witness layout and replays every minor certificate. It reports no failure on the 16,949 records. A value certified by search cannot be checked from its record; it rests on the repaired search, and for instances within reach of the emitter the certificate of Section 5.3 is how a third party can check it.

Agreement with published values. All 84 Small graphs agree with the pathwidths of Mallach (2018, Tables 4–5). Ten of the eleven VLSI circuits are certified at their published best-known track counts (de Oliveira and Lorena, 2002, Table I); the eleventh, W4 with 202 nets, is open at 28 against 27.

A caution about MOSP files. A MOSP file can be read with rows as customers or with rows as patterns, and the two readings give different graphs. The same public repository ships the de Carvalho and Soma (2015) instances as customers by patterns and the Chu and Stuckey (2009) instances beside them as patterns by customers. Only published densities and values tell them apart. Read the wrong way round, an instance certifies at a different value.

Table 5.3: Twenty collections, 21,754 instance files, reduce to 17,714 isomorphism classes, and 16,087 of them have a certified pathwidth. The open classes are mostly in the PACE sets, the VSPLIB grids and Harwell-Boeing graphs, and the MOSP sets of Carvalho and Soma and of Frinhani et al. *Classes*: isomorphism classes within the collection (78 graphs above 5,000 vertices are compared by file content only). *Counted here*: classes whose first member, in the order of the table, is in this collection, so the column sums to the dataset’s class count. The next four columns split those classes by provenance; *No witnessed value*: no layout is stored yet. ^aThe same 46 files distributed a second time under other names; every class is counted under its first copy.

Collection	Files	Classes	Counted here	Certified		Upper bound only	No witnessed value
				by search	by bound		
MOSP: Challenge 2005	5,806	3,169	3,169	3,168	1	0	0
MOSP: Challenge 2005 (second copy) ^a	46	46	0	0	0	0	0
MOSP: Chu & Stuckey	200	200	200	198	0	2	0
MOSP: Faggioli & Bentivoglio	300	287	274	274	0	0	0
MOSP: SCOOP	24	24	24	24	0	0	0
VLSI gate matrix circuits	11	11	11	5	5	1	0
VSPLIB trees	50	28	28	28	0	0	0
VSPLIB grids	50	50	50	9	0	19	22
VSPLIB Harwell-Boeing	73	72	72	32	7	33	0
Small (Marti et al. 2008)	84	84	84	32	52	0	0
Rome graphs	11,534	11,199	11,199	8,126	2,746	326	1
TreewidthLIB colouring	58	58	58	18	13	27	0
freetdi named graphs	150	146	143	97	22	23	1
freetdi control-flow graphs	1,817	1,070	1,064	398	661	4	1
MOSP: Frinhani et al. 2018, large	610	610	610	0	0	0	610
PACE 2016	291	288	112	37	18	51	6
PACE 2017 exact	200	199	193	61	7	115	10
PACE 2017 heuristic	200	197	173	11	3	45	114
PACE 2017 bonus	100	100	100	23	0	77	0
MOSP: Carvalho & Soma 2015	150	150	150	11	0	139	0
All	21,754		17,714	12,552	3,535	862	765

6 Conclusion

Placeholder. Section 6 is written last, once Sections 1–5 are final. It is to say what the complex is, what is proved and where, and what remains open: the missing Table 1 reference (Kashiwabara and Fujisawa 1979); the Table 1 claims that turned out false or only bands; LaPaugh’s $es = pes$, stated and not formalised; the two never-certified values still in flight (Section 5.2); the price of the open dataset classes (moved from Section 5.4, see the comment below); the certificate’s acceptance condition, argued on paper and not in Lean; and the gap between pathwidth and treewidth on trees of cliques, where no degree or clique bound can close it.

Data, code and proofs

Everything this paper reports can be checked from its repository, URL to be added on release. The repository holds:

- the Lean 4 development, with every theorem the paper names (Appendix A) and an axiom check that covers all of them;
- the source of both solvers, the open-stacks customer search and the graph pathwidth solver, in C and in Python, with the published and the repaired rules;
- the dataset of Section 5.4, the certificates of Section 5.3 and the independent checkers of both;

- the bibliometric data of Section 3: the cached OpenAlex responses, the relevance labels and the mechanical rule;
- a script for every table, figure and number in the paper, each of which regenerates it from the recorded data.

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A Formal names

Every theorem, lemma and counterexample of this paper is checked in the Lean 4 proof assistant with the Mathlib library, unless the text says otherwise. The reader need not read Lean to follow the paper. This appendix is for a reader who wants to find the formal statement of a result: Table A.1 maps each result, in the order in which it appears, to the names of the Lean declarations that state it. Where a row cites a definition rather than a theorem, it is the definition the text uses.

All the theorems named depend only on the three standard axioms of Lean’s classical logic, `propext`, `Classical.choice` and `Quot.sound`; an axiom check in the repository prints this for every theorem named in the paper. The development has one `sorry`, a conjecture about lower bounds that no result of this paper uses. The part of the development behind Section 4 has no `sorry` at all.

Two Lean proofs take a route that the text does not describe. The proof of Theorem 4.2 goes through narrowness (Kornai and Tuza, 1992), which counts the closing order forwards rather than reversing it; and the proof of Theorem 4.7 reduces the deficiency form of Hall’s theorem to Mathlib’s Hall theorem by adding δ dummy stacks. One Lean proof takes a different route from the published one: the proof that node search is monotone (Theorem 2.4(5)) follows the crusade argument of Bienstock and Seymour (1991), not the reduction to edge search of LaPaugh (1993) on which Kirousis and Papadimitriou (1986) rely. A few facts in the paper are computed rather than proved: that every end of the two bands occurs (the values of K_2 , $K_{1,3}$ and $K_{3,3}$ after Theorem 2.5), each computed by brute force from the quantity’s own definition; the search for smaller counterexamples (Appendix B.5); and the computational results of Section 5.

Table A.1: Each result of the paper and its formal statement in Lean.

Entry	Result in the paper	Lean names
<i>Section 2, the equivalences</i>		
A.1	Lemma 2.1, cliques sit in a bag	PathDecomposition.exists_bag_of_isClique
A.2	Theorem 2.2, vertex separation is pathwidth	vertexSeparation_eq_pathwidth
A.3	Theorem 2.3, MOSP is pathwidth plus one	MOSPInstance.mospValue_eq_pathwidth_add_one
A.4	The pattern graph gives no bound: the star $K_{1,4}$ (after Theorem 2.3)	star_mospValue, star_pathwidth_agreementGraph, star_refutes_pattern_graph_bound
A.5	Theorem 2.4(1), gate matrix layout	NetGateMatrix.tracks_eq_mospValue, NetGateMatrix.tracks_eq_pathwidth_add_one
A.6	Theorem 2.4(2), interval thickness	intervalThickness_eq_pathwidth_add_one
A.7	Theorem 2.4(3), one-dimensional logic	LogicArray.tracks_eq_intervalThickness, LogicArray.tracks_eq_pathwidth_add_one
A.8	Theorem 2.4(4), narrowness	narrowness_eq_pathwidth_add_one
A.9	Theorem 2.4(5), node search	nodeSearchMonotonicity, nodeSearch_eq_vertexSeparation_add_one, nodeSearch_chain
A.10	Theorem 2.4(6), multiple folding	NetGateMatrix.foldTracks_eq_pathwidth_add_one
A.11	Theorem 2.4(7), Lengauer’s vertex separator game	vsg_eq_max
A.12	Theorem 2.5, the bands	pathwidth_le_splitBandwidth_le_pathwidth_add_one, vertexSeparation_le_edgeSearch_le_add_two, vertexSeparation_le_progressiveEdgeSearch_le_add_two
A.13	Theorem 2.6(1), PLA folding	plaTracks_idMatrix_five, plaTracks_idMatrix_unbounded, plaTracks_pathMatrix_unbounded
A.14	Theorem 2.6(2), edge separation read as cutwidth	cutwidth_unbounded, modCutwidth_unbounded
<i>Section 4, the search</i>		
A.15	Step cost, P_k and Sol_k (Section 4.1)	stepCost, Solvable, SearchSol
A.16	Lemma 4.1, free moves never hurt	solvable_iff_searchSol_cl
A.17	Theorem 4.2, the search decides pathwidth	searchSol_empty_iff_pathwidth_add_one_le, searchSol_mospGraph_iff_mospValue_le
A.18	Table 4.1: closing order from $\emptyset$	orderOfLayout
A.19	Table 4.1: open stacks $b(T)$	openStacks_eq_card_boundary
A.20	Table 4.1: $cost(T, c)$	stepCost_eq_card_boundary_add_one
A.21	Table 4.1: $Sol_k(\emptyset)$	searchSol_empty_iff_pathwidth_add_one_le
A.22	Table 4.1: free closure $cl(T)$	mem_cl_iff
A.23	Table 4.1: definite move, repaired	isHereditarilyDefinite_iff_isCommittable
A.24	Lemma 4.3, what the premise says	closeCount_eq, isDefinite_iff

Entry	Result in the paper	Lean names
A.25	Counterexample 4.4	definiteMove_counterexample, cexFamily_closed
A.26	Chu and Stuckey’s Theorem 1, under both readings, and Chu’s Theorem 6.3.6 are false	chuStuckey_theorem1_false, chuStuckey_theorem1_false_literal, chuThesis_theorem636_false
A.27	Fink’s restatement of the definite move is false	fink_theorem1_false
A.28	The hereditarily definite premise, and its failure in Counterexample 4.4	IsHereditarilyDefinite, not_isHereditarilyDefinite_cex
A.29	Lemma 4.5, open stacks are submodular	openStacks_submodular
A.30	Theorem 4.6, the repaired definite move is sound	solvable_cl_insert_of_ hereditarilyDefinite, searchSol_cl_ insert_of_hereditarilyDefinite
A.31	Theorem 4.7, the repair is a matching condition	isHereditarilyDefinite_iff_ hasDefiniteMatching
A.32	The published definite move is right when $\text{open}(q, S) \leq 1$	isHereditarilyDefinite_of_openCount_ le_one
A.33	The subset rule is sound	searchSol_cl_insert_of_newlyOpened_ subset, subsetFilter_sound
A.34	The subset rule needs its tie-break (7-customer graph)	noTieBreak_counterexample
A.35	The better move’s premise, restated	isBetter_iff
A.36	Chu and Stuckey’s Theorem 2 (the better move) is false	betterMove_counterexample, chuStuckey_theorem2_false
A.37	Chu’s Theorem 6.3.8 is false	chuThesis_theorem638_false, chuThesis_theorem638_false_literal
A.38	The repaired better move is sound	IsRepairedBetter, searchSol_cl_insert_of_repairedBetter
A.39	Two implementation faults of the better move	bugA_counterexample, bugB_counterexample
A.40	The old move is sound	searchSol_reinsert, not_searchSol_of_oldMove
A.41	The old move needs its reinsertion test (path on five vertices)	reinsert_needs_test
A.42	Solvability of a state does not depend on the path to it (the memo)	solvable_iff_of_cl_eq
A.43	A node with an unrefuted old move can record a solvable state	exec_fake_oldMove
A.44	Node-soundness of a filter (Section 4.5)	NodeSoundAt
A.45	Theorem 4.8, the revised search is sound	exec_repairedFullFilter_sound, exec_repairedFullFilter_pathwidth, exec_repairedFullFilter_mospValue
A.46	Under the published rules the filter is not node-sound	not_codeFilterSound_cexGraph
A.47	Lemma 4.9, the premises as commitment conditions	isDefinite_iff_endpoint, isHereditarilyDefinite_iff_ isCommittable
A.48	In Counterexample 4.4 the endpoint test passes and the commitment fails	cex_isDefinite_not_isCommittable
A.49	The published premise gives a commitment to the least-border set	exists_isCommittable_of_isDefinite
A.50	In Counterexample 4.4 that set is $\{2, 3, 4\}$	cex_isCommittable_234

Entry	Result in the paper	Lean names
A.51	A commitment of depth one holds iff $\text{open}(q, S) \leq 1$	<code>isCommittable_insert_iff</code>
A.52	The greedy step of Coudert, Mazauric and Nisse is that condition	<code>isGreedyStep_iff</code>
<i>Section 5, computational results</i>		
A.53	The split of the root into independent tasks is sound	<code>execSplit_repairedFullFilter_sound</code> , <code>execSplit_repairedFullFilter_mospValue</code>
<i>Appendix B, supporting material</i>		
A.54	Node search equalities need an edge (Appendix B.2)	<code>nodeSearch_ne_intervalThickness_of_edgeless</code> , <code>monotoneNodeSearch_ne_vertexSeparation_add_one_of_edgeless</code>
A.55	Lengauer’s Theorem 4 fails at $K = 0$ (Appendix B.2)	<code>isPositiveVSG_triangleGraph_counterexample</code>
A.56	$t = \text{ns}$ fails on the matrix [1] (Appendix B.2)	<code>tracks_ne_monotoneNodeSearch_one</code>
A.57	The Kirousis–Papadimitriou claim fails on $K_{1,3}$ (Appendix B.2)	<code>kirousisPapadimitriou_claim2_false</code>
A.58	Progressive pebbling is in the exact core (Appendix B.3)	<code>pbw_lengauerD_eq_pathwidth</code> , <code>pbw_eq_mospValue_pebbleMatrix</code> , <code>mpb_eq_pathwidth_add_one</code>
A.59	Proposition B.1, the repaired filter is node-sound	<code>repairedFullFilter_sound</code>
A.60	Theorem B.2, runs are sound	<code>Exec.sound</code>

B Supporting material

This appendix holds the material that supports Sections 2, 4 and 5 without being needed to follow them.

B.1 Formal definitions of eight problems

These are the formal definitions of the eight problems that Table 2.1 states in one line. As in Section 2.1, each is a decision problem with an integer k , and its value is the least k for which the answer is yes.

One-dimensional logic. Given gates, nets and the nets each gate connects, is there an order of the gates whose net intervals have an interval graph of clique number at most k ? Its *net graph* H joins two nets that share a gate. A *boundary* variant pins two gates to the ends (Ohtsuki et al., 1979, §§II–IV).

PLA folding $\text{pla}(M)$. As gate matrix layout, with at most two nets per track. With no such bound (*multiple folding*) it is gate matrix layout itself (Möhring, 1990, pp. 25, 36).

Interval thickness $\theta(G)$. Is there an interval supergraph of G on the same vertices with clique number at most k ? (Kirousis and Papadimitriou, 1985, p. 182); (Möhring, 1990, p. 31).

Node search $\text{ns}(G)$. Can searchers, placed on and removed from vertices, clear every edge with at most k on the graph at once, where an edge is cleared when both ends are guarded and a clear edge is recontaminated by a searcher-free path to a contaminated one? $\text{mns}(G)$ is the same over strategies that never recontaminate (Kirousis and Papadimitriou, 1985, p. 181); (Kirousis and Papadimitriou, 1986, §2).

Edge search $es(G)$. The same game with a third move, sliding a searcher along an edge, which clears that edge; $pes(G)$ for progressive strategies, which never recontaminate an edge (Kirousis and Papadimitriou, 1986, p. 208); (Ellis et al., 1994, p. 53).

Narrowness $v(G)$. Is there an order $v_1, \dots, v_n$ such that the following process never holds more than k vertices: at step i add v_i , then remove every v_j , $j \leq i$, with no neighbour v_ℓ , $\ell > i$? (Kornai and Tuza, 1992, §2).

Split bandwidth $sb(G)$. Is there a graph obtained from G by repeatedly splitting a vertex into an edge, its neighbours partitioned between the two ends, with bandwidth at most k ? (Fomin, 1998, §3.2).

Edge separation. The Linhares–Yanasse table cites Lengauer (1981), which supports three readings: cutwidth $cw(G)$, the least over layouts of the most edges crossing a gap (p. 468); modified cutwidth $mcw(G)$, counting only edges that pass strictly over a vertex (Definition 6); and Lengauer’s vertex separator game $VSG(G)$, the least K such that the vertices can be pebbled in an order that leaves at most K unpebbled vertices adjacent to pebbled ones (p. 467).

B.2 Edge cases and a gap in a published proof

Three of the published equalities fail on degenerate inputs, and each failure is proved in Lean. On a nonempty graph with no edges, $ns = 0$ while $vs + 1 = \theta = 1$, so both equalities of Kirousis and Papadimitriou (1985, 1986) need an edge (Appendix A, A.54). For a matrix with no 1s, $Z = 0$ while $pw + 1 = 1$. Lengauer’s Theorem 4 fails at $K = 0$ on an edgeless graph (Appendix A, A.55). Möhring (1990, Thm 3.9), $t = ns$, fails literally on the 1×1 matrix $[1]$ (Appendix A, A.56).

The published proof of $vs \leq ns - 1$ (Kirousis and Papadimitriou, 1986, Thm 4.1) has a gap. It orders the vertices by when they first receive a searcher and claims that every vertex counted at the worst cut still holds a searcher, “since the vertices in D_{i_0} are joined to vertices not yet cleared” (p. 217). A strategy may place a searcher and remove it again before clearing anything, which recontaminates nothing. On $K_{1,3}$, placing and removing a searcher on each leaf before guarding the centre is an optimal recontamination-free strategy for which the claim fails (Appendix A, A.57). The theorem is true. The repair orders the vertices by clearing time, as our Lean proof does, or uses the paper’s own normal form, its Corollary 2.4.

B.3 A thirteenth member

Progressive black-white pebbling, which is not in the Linhares–Yanasse table, belongs to the exact core. Lengauer (1981, Thm 3) gives $vs(G) \leq K \iff pbw(G_d) \leq K + 2$ on the dag G_d built from G ; his Theorem 2 gives $pbw(D) = vs(D_u) + 1$ on a dag D , where D_u is the MOSP graph of the matrix whose column v is v with its predecessors; and the minimum progressive pebbling number of Kirousis and Papadimitriou (1986, Thm 3.1) equals $pw + 1$. All three are proved in Lean, each with its hypothesis: Lengauer’s Theorem 2 in its instance form fails at $K = 1$ on a dag with no arcs, and $mpb = ns$ fails on edgeless graphs (Appendix A, A.58).

B.4 Why the published proof of the definite move fails

The proof moves q to the front of a solution and claims that after each prefix “we have at most $open(q, S)$ extra stacks open, but at least $close(q, S)$ extra stacks closed” (p. 6). At a prefix $T \supseteq S$ that does not yet contain q , moving q forward changes the open stacks by $b(T \cup X) - b(T) = |N[q] \setminus O(T)| - |X \setminus T|$. At $T = S$ this is $open - close \leq 0$. But customers of X that the solution has *already* closed are in T and no longer count as extra stacks closed, while the stacks q opens may still be new. In the example, a solution

may begin with 3, after which 4 is free: at $T = \{2, 3, 4\}$, moving 0 forward costs one more stack. A second, independent slip: even where the conclusion holds, the literal sequence $S ++ [q] ++ R'$ need not be a solution, because a customer that q frees stays open until its turn; the argument needs the free moves.

B.5 The search for smaller counterexamples

The smallest graph on which the rule loses the last solution has 14 customers: there is none on any graph to 9 vertices, by exhaustion, and none on the 88,802 connected graphs on 10 vertices with at most 14 edges. The counterexample was found by a search over a family of gadgets built from the failed proof, and minimised by deletion.

B.6 The search for prior reports of the error

We looked for any earlier report of the error in the definite move. We listed the works citing [Chu and Stuckey \(2009\)](#), Chu’s thesis and its IJCAI 2013 abstract in OpenAlex, Semantic Scholar and OpenCitations: 141 works. Of the 38 that cite the CP paper we read 24 in full (63%), and of the 103 that cite only the thesis, 32 (31%). No work read gives a counterexample, an erratum or a correction. One, [Fink \(2012\)](#), restates the definite move as true with a stronger premise, refuted in Section 4.3. Three use an implementation of the rules and report no failure; among them [Frinhani et al. \(2018\)](#) ran the original code to obtain the optima that [Martin et al. \(2022\)](#) reuse.

B.7 The filter and the runs are sound

Proposition B.1. *At every state with the invariant, for every set Q of refuted old moves, the filter definite $\rightarrow$ subset $\rightarrow$ better with both repairs is node-sound (Appendix A, A.59).*

Proof. If the definite move fires, Theorem 4.6 applies. Otherwise the subset rule keeps a survivor with a solution; let w be the earliest. If the better move dropped w , it cited an earlier survivor that the repaired argument proves solvable, contradicting the choice of w . $\square$

Theorem B.2 (runs are sound). *Let the filter be node-sound and return only playable customers. If at the start of a run the invariant holds, every memo entry is a genuine refutation, and every old move is refuted, then the run leaves only genuine memo entries, and if it answers false at S then $\neg \text{Sol}_k(S)$. This holds for any child order and any memo policy, with the old move on or off (Appendix A, A.60).*

The proof is an induction over the run, in the order in which subtrees complete: every old move a node holds is a sibling subtree that completed with *false* earlier in the same run, or an inherited one that passed the reinsertion test.

B.8 The split re-certification

Table B.1 gives the seven hardest re-certifications of Section 5.2 and the two values never certified, which the same split is refuting. Each instance’s tree was expanded at the top into independent subtrees (tasks), and each task was refuted by one worker. The seven re-certifications ended with every task refuted and the value unchanged. No task of any instance has found an order of the smaller value.

Table B.1: All seven hardest values re-certify unchanged, at 18 to 369 core-hours each; the two values never certified are in flight, with no task satisfiable so far (read 6 October 2026). *Value*: the stored optimum; *k*: the value minus one, which must be refuted. For an instance in flight, the nodes and core-hours are those of the tasks finished so far.

Instance	Value	<i>k</i>	State	Tasks refuted	Nodes	Core-hours
Random-125-125-4-1_0	57	56	refuted	669	4.97×10^{10}	18.1
Random-125-125-2-4_0	24	23	refuted	1,686	4.98×10^{11}	154.1
Random-125-125-4-5_0	46	45	refuted	616	7.96×10^{10}	26.1
Random-125-125-4-2_0	57	56	refuted	842	7.64×10^{10}	30.2
Random-125-125-2-1_0	24	23	refuted	4,096	1.30×10^{12}	366.5
Random-125-125-2-5_0	20	19	refuted	4,168	1.28×10^{12}	368.8
Random-125-125-4-4_0	51	50	refuted	2,504	4.12×10^{11}	148.8
Random-125-125-2-3_0	21	20	in flight	2,354	1.19×10^{12}	434.3
Random-125-125-2-2_0	25	24	in flight	2,141	1.02×10^{12}	431.0

C The bibliometric study

This appendix gives the method behind Section 3 in full, the counts per name and when each name was in use.

Search. We measured how much each name is used with OpenAlex (Priem et al., 2022), on 29 September 2026. For each of the twelve problems we searched titles and abstracts for its name and its common variants, and restricted the search to the four fields in which the problems are studied: Computer Science, Mathematics, Engineering and Decision Sciences. Without that restriction several names are ordinary words: “narrowness” returns over a million works, and “edge separation” returns aerodynamics.

Relevance. Inside those fields a search hit is still not a work that uses the name. OpenAlex’s search stems words and ignores punctuation, so a quoted phrase matches more than the phrase. “Narrowness” matches “narrowing” in term rewriting; “node searching” matches look-ups in peer-to-peer networks; “split bandwidth” matches radar interferometry and 5G spectrum assignment; “path-width” matches the widths of roads, tool paths and datapaths. We therefore fetched all 2,271 hits with their titles, abstracts, topics and venues, and kept a work only if it is about the problem in the sense of the Linhares–Yanasse table. For the eleven smaller names the 700 hits were judged one by one. For pathwidth, the 1,571 hits were filtered by a rule: keep a work that uses the one-word spelling, that OpenAlex places in a theory subfield, or that mentions graphs, treewidth or minors and not roads, vehicles, trajectories or datapaths. We checked the rule by reading samples of what it kept and what it dropped. Of the 2,271 hits, 1,593 are relevant: 1,213 for pathwidth and 380 for the other eleven names. The two search-game names keep 27% and 40% of their hits.

Robustness. The counts are lower bounds. A work can study a problem without naming it in its abstract, OpenAlex has few abstracts before about 1990, and its citation counts run below Google Scholar’s. The judgements were made by one annotator with model assistance, record a yes or no per work and not the reason, and have not been checked by a second annotator. As a second reading we applied a mechanical rule: the literal phrase with a context term and no exclusion term. It agrees with the judgements on 543 of the 700 judged works (78%). Under both readings pathwidth has more than three times as many works as the other eleven names together (3.2 and 3.6 times), the three empty names stay empty apart from one 5G false positive, and the search games lose most of their hits. MOSP’s rank over all years is the one conclusion that depends on the reading: fourth under the judgements, fifth under the rule.

Table C.1: Works that use each problem’s name in the sense of the Linhares–Yanasse table, and citations of the paper the table cites for it. *Relevant*: hits kept by the relevance check. *Citations*: OpenAlex `cited_by_count`, 29 September 2026. Kashiwabara and Fujisawa (1979) has no OpenAlex record. “OpenAlex returned one pathwidth record twice: the counts are of records, as Figure 3.1 draws them, and there are 1,212 distinct relevant pathwidth works.

Problem	Discipline	Relevant	Hits	Reference in the 2002 table	Citations
Pathwidth	graph theory	1,213 ^a	1,571	Kinnersley (1992)	214
Gate matrix layout	VLSI	110	125	Möhring (1990) ; Wing et al. (1985)	134 / 89
PLA folding	VLSI	68	74	Möhring (1990)	134
MOSP	operations research	58	60	Yanasse (1997b) ; Fink and Voß (1999)	75 / 71
Vertex separation	graph theory	55	100	Kinnersley (1992)	214
Edge search	graph theory	38	95	Kirkousis and Papadimitriou (1986)	293
Node search	graph theory	34	127	Kirkousis and Papadimitriou (1985)	124
One-dimensional logic	VLSI	11	18	Ohtsuki et al. (1979)	108
Interval thickness	graph theory	6	10	Kashiwabara and Fujisawa (1979)	–
Narrowness	graph theory	0	35	Kornai and Tuza (1992)	44
Split bandwidth	graph theory	0	35	Fomin (1998)	19
Edge separation	graph theory	0	21	Lengauer (1981)	77
All twelve		1,593	2,271		

Counts and citations. Table C.1 gives the counts per name. Citation counts do not track name usage. [Kirkousis and Papadimitriou \(1986\)](#) is the most-cited paper in the table (293 citations), while only 38 works use “edge search” in its sense: the paper is cited as a foundational result on graph searching, not because people work on the edge search game under that name. The split is starker for [Ohtsuki et al. \(1979\)](#): 108 citations against 11 works using “one-dimensional logic”. Split bandwidth is Fomin’s own term in the paper the table cites ([Fomin, 1998](#)), and no one adopted it.

When each name was in use. Figure C.1 shows the counts over time. PLA folding and gate matrix layout peak in 1980–1989 and fall to almost nothing after 2010, and one-dimensional logic has no relevant work after 2009. MOSP first appears in the 1990s, peaks in 2005–2014 and is still active. Pathwidth’s rate has roughly doubled since the 1990s, from about 15 to 27 works per million, and its 311 relevant works in 2020–2024 are more than the whole history of any other name. Rows with a few works per period move by whole cells on one paper, and should be read as anecdote rather than trend.

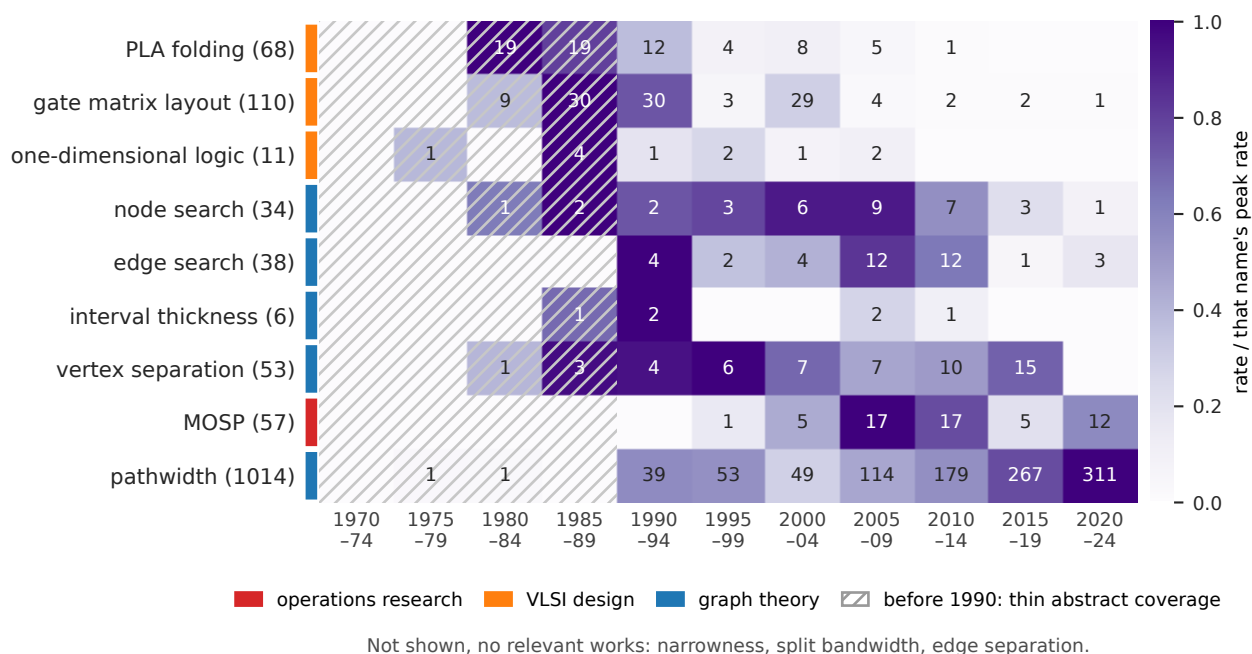


Figure C.1: The VLSI names belong to the 1980s, the search games peak later and fade, MOSP peaks in 2005–2014, and only pathwidth is still rising. The shade is the name’s rate, relevant works per million works in the same four fields in each five-year period, as a fraction of that name’s own peak rate; the numbers are raw counts. Rows are ordered by the period of their peak. Counts cover 1970–2024, since 2025–2026 is not fully indexed, so row totals are below those of Table C.1. Before 1990 (hatched) OpenAlex has few abstracts.